MIND AI × ONTYPOLOGY

AI Bio Lab Brain

Output the complete protocol of the correct method



Space Type × Time Type

On the Mind AI base platform | Type = Type

Worked case | Enzyme kinetics assay: Km · Vmax

Biotech R&D lab edition

Ontypology × Mind AI solution series

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Foreword: Three Conventions

Welcome to this methodology document. It has exactly one task: to make every output of the AI Bio-Lab Brain — hypothesis, experimental design, SOP, data, verdict, evolution — accurate, acceptable and traceable. The book is grounded in Ontypology (Space Type + Time Type) and runs on the Mind AI platform; the running case throughout is the most ordinary task in a biotech R&D lab: establishing a kinetic specification for an enzyme (alkaline phosphatase, ALP, with pNPP as substrate, solving for Km and Vmax).

> **Three Conventions (Ground Rules Before You Read)**
>
> Convention One — Type, never Form: throughout the book every Output Type is named with Type, as in HypothesisType, ExperimentType and ProtocolType.
>
> Convention Two — the theory is used, not taught: the two Types of Ontypology (Space Type and Time Type) enter only as tools. The three bracket types, ∩(T), id(C), the remove criterion, the four-step framing and the move board are used directly, without restating the original arguments and chapters.
>
> Convention Three — the platform is cited, not described: Mind AI is the underlying platform, so the division of labour between the LLM and the Mind Model is cited only, not restated from existing Mind AI material.

Statement on Illustrative Data (Said Up Front)

All experimental data, SOP numbers, batch numbers and people in this book (Engineer Wu, the new rALP batch) are illustrative. They exist to demonstrate output formats and reasoning paths; they do not represent real experimental results and must not be used as a basis for clinical, regulatory or commercial decisions.

Biochemical work involves reagents, instruments and biological materials; carry it out in accordance with each laboratory's safety rules and applicable regulations. This book involves no system implementation; it delivers only the method for producing accurate output, together with the full content of the proposal.

Table 0-1 Task Map: Six Chapters, One Road

Chapter	Theme	Emphasis	In One Line
Foreword	Three conventions and the illustrative-data statement	Set the ground rules first	Agree how to read before starting

Chapter	Theme	Emphasis	In One Line
Introduction	Two languages, three ways to read	Comparison table; reading paths	See the map before setting off
Chapter 1 (System Positioning)	What this brain is	The experimental twin; where the two Types land; division of labour	What it is, and what makes it accurate
Chapter 2 (Output Skeleton)	Overview of the six Output Types	Hypothesis, experiment, protocol, data, verdict, evolution, taken one by one	What the outputs look like
Chapter 3 (Accuracy Gatekeeping)	The four accuracy gates	Criterion; skeleton; validation; evidence chain	Where accuracy comes from
Chapter 4 (Operating Procedure)	From requirement to experiment	Four-step framing; six steps; three-layer judgement	How to do it in practice
Chapter 5 (Prompt Examples)	S1–S7 and a full-run example	Full prompts; the ALP kinetics case	Copy it and use it
Chapter 6 (Governance and Adoption)	Misuse comparison and maturity	Five kinds of misuse; L1–L4; the blueprint	How to make it last
Review (Closing)	Three things to take away	The twin; the six-Type chain; the four layers	What to carry away
Appendix A (Quick Reference)	Terms, moves, notation	Three quick-reference tables	Kept for lookup

Introduction: Two Languages, Three Ways to Read

This book joins two languages: the language of Type in Ontypology, and the language of doing experiments in the lab. Before setting off, take one look at the whole picture — Space Type describes what something is, Time Type describes how it is done, and Mind AI loads both into a single experimental twin; what runs out of that twin is the six Output Types.

Figure D-1 Dual-Type panorama: the Space Type depicts "what it is", the Time Type depicts "how to do it"

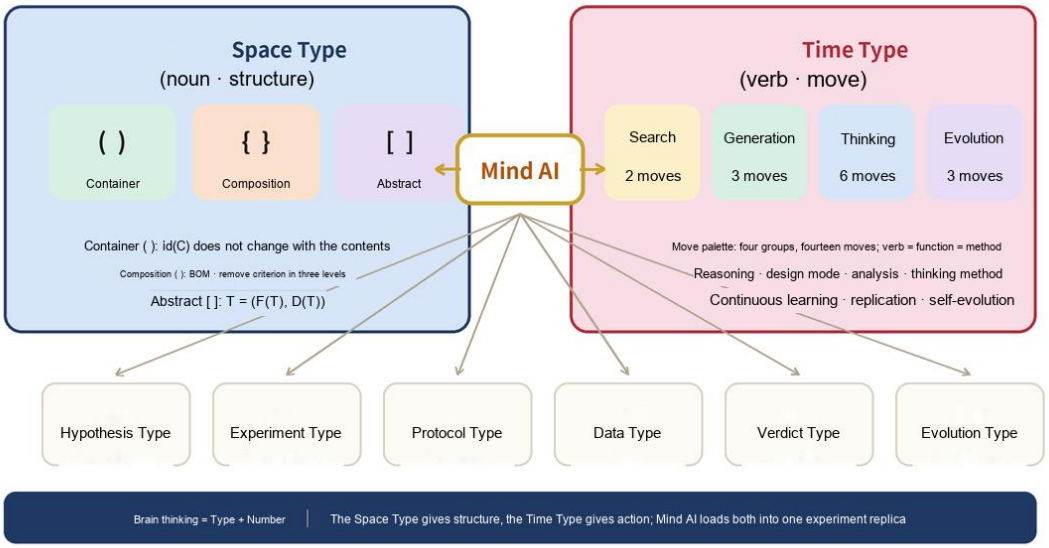


Figure D-1 The Two Types in Panorama: Space Type describes what something is, Time Type describes how it is done

Table D-1 Two Languages Side by Side: Type terminology on the left, bench language on the right

Ontypology tool (used, not taught)	Bio-lab counterpart	In One Line
Container ( ) | C = (id(C), S(C))	Well, tube, petri dish, sample tube	Identity does not change with the contents
Composition { } | BOM; remove in three levels	Reaction recipe, medium, reagent composition	Take it apart — does it still know itself?
Abstract [ ] | T = (F(T), ∩(T))	Assay category, specification, acceptance criterion	Every feature must pass
Four-step framing	Fixing the requirement; positioning before the hypothesis opens	A failed control forces out a new dimension
Three-layer judgement	Assay eligibility check (is / has / right)	Ask Container, BOM and Abstract once each
Time Type; the fourteen moves of the move board	SOP actions, analytical techniques, learning and evolution	Verb = function = method = move
∩(T) intersection criterion	The four data-quality gates	No gate passed, no next step
Type evidence chain	Tracing back through the lab record	Every conclusion traces to its source

Table D-2 Three Ways to Read: Different roles, different paths

Role	Suggested path	What you can do afterwards
Lab manager	Introduction → Chapter 1 (System Positioning) → Chapter 6 (Governance and Adoption) → Review	Grasp the positioning, the risks and the pace of adoption

Role	Suggested path	What you can do afterwards
Principal investigator (PI)	Chapter 1 (System Positioning) → Chapters 2 and 3 (Skeleton and Gates) → 5.3–5.5	Build habits of accuracy acceptance and review
R&D engineer	Chapter 4 (Operating Procedure) → Chapter 5 (Prompt Examples) → Appendix A	Go straight to S1–S7 and the move board

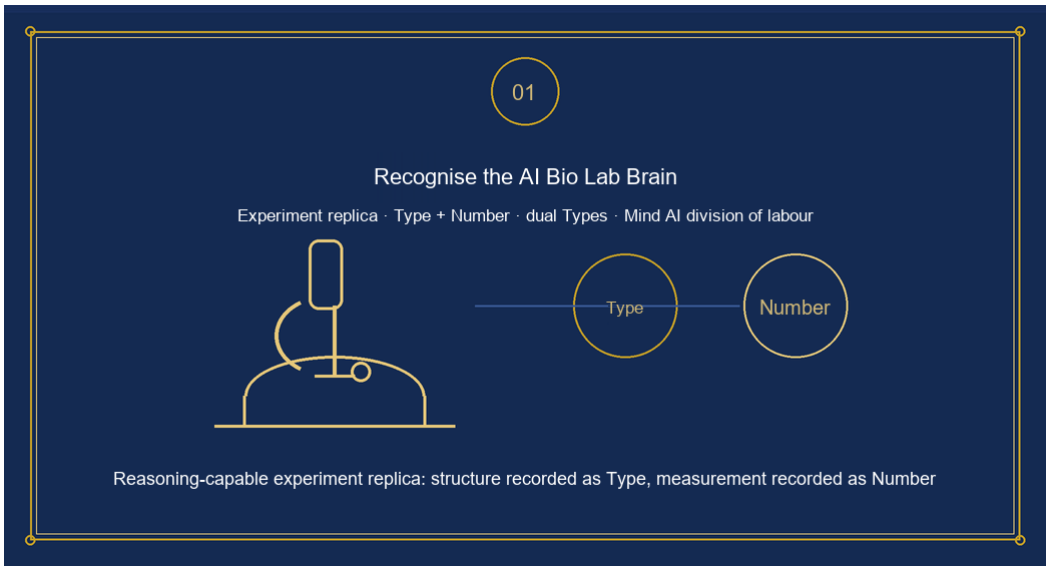
Warm-Up: Three Sentences to Get Started

First: to do an experiment accurately, build the Type correctly first — get the structure definition wrong, and the prettiest data is worth nothing.

Second: accurate output has to be acceptable — agree on ∩(T) up front, and a dispute has a referee.

Third: experimentation is an art of verbs — the move board of Time Type turns how to do something into moves you can pick, combine and swap.

1 Understanding the AI Bio-Lab Brain



Before handing over any output, settle three things: what this brain contains (a reasoning-capable experimental twin), how the theory lands on the bench (the two Types), and who is responsible for guarding accuracy (the division of labour in Mind AI).

1.1 The Reasoning-Capable Experimental Twin: Type + Number

The most valuable thing in a laboratory is not the spectrophotometer; it is the collective memory of what was done, what was seen and why that conclusion followed. The AI Bio-Lab Brain builds that memory into a reasoning-capable experimental twin: structure is recorded as Type — what Container the sample sits in, what the recipe is Composed of, which class the assay belongs to; measurement is recorded as number — concentration, absorbance, initial rate, statistics. Type gives the skeleton, number gives the flesh, and only then can the twin reason.

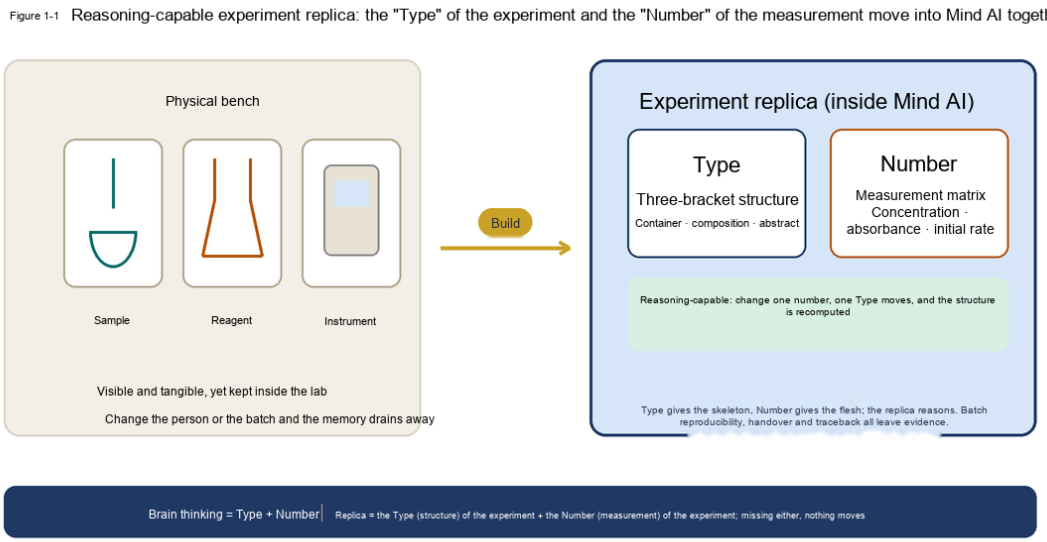


Figure 1-1 The Reasoning-Capable Experimental Twin: the bench's Types and the measurement's numbers move into Mind AI together

The value of the twin is practical: change the person, change the batch, come back six months later — the reasoning path is still there. Comparing batches no longer rests on a feeling that it is about the same as last time; you take out the same set of Types and the same set of criteria and recompute.

Table 1-1 Four Scenes of Inaccurate Output: you have met them all

Scene	Symptom	How the twin solves it
Inconsistent reporting conventions	One dataset, two people, two verdicts	Set ∩(T) in the Verdict Type first; agree on the referee
Protocols passed on by hand	The SOP lives with the senior technician; change hands and it drifts	Version the Protocol Type; every action is a move
Data with no traceability	Press a conclusion and you get 'I think it was something like...'	The Type evidence chain: all six Types checkable link by link
Specifications by feel	'About the same as the last batch', with no number attached	The reasoning skeleton plus a CI-based equivalence criterion

1.2 The Two Types Land on the Bench

Space Type and Time Type each hold a post in the lab. Space Type governs what a sample is: a tube is at once a Container (( ): the well keeps its identity), has a Composition ({ }: the reaction recipe) and belongs to a class ([ ]: the criterion for its assay category). Time Type governs how the experiment is done: verb = function = method = move. The fourteen moves fall into four groups — Search, Generation, Thinking, Evolution — and every action in

an SOP and every technique used in analysis maps onto one of them.

Figure 1-2 The dual Types in practice: the Space Type governs "what it is", the Time Type governs "how the experiment is done"

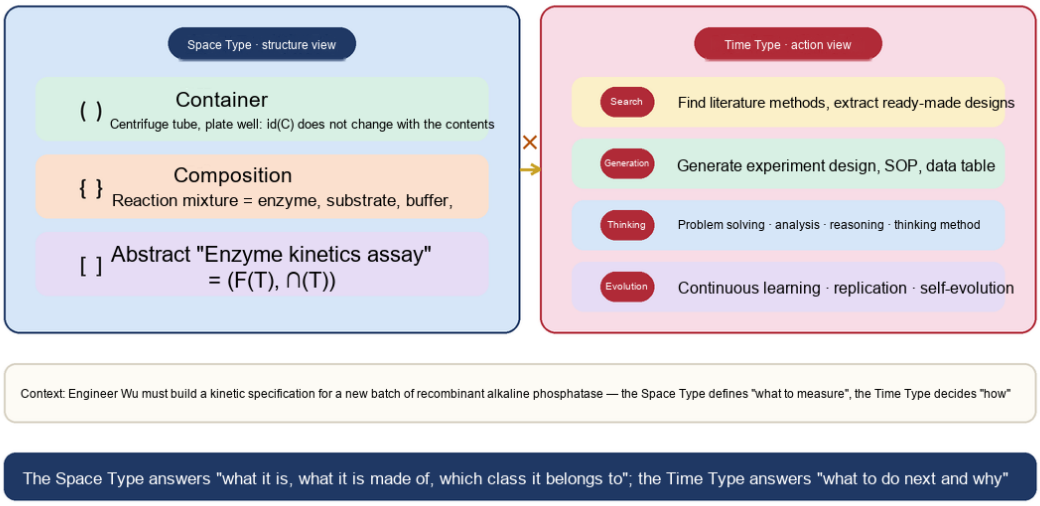


Figure 1-2 The Two Types Land on the Bench: Space Type governs what a sample is, Time Type governs how the experiment is done

Table 1-2 The Three Space Brackets × Where They Land on the Bench

Bracket	Type tool	Bench landing point	Question asked
( ) Container	C = (id(C), S(C)); the invariance axiom	Well, tube, petri dish, cryovial	Does identity hold?
{ } Composition	BOM; the remove criterion in three levels	Reaction recipe, medium, reagent composition	Does it come apart?
[ ] Abstract	T = (F(T), ∩(T))	Assay category, specification, acceptance criterion	Do all features pass?

Table 1-3 The Move Board: Four Groups, Fourteen Moves × Where They Land in the Lab (full table in Appendix A)

Group	Moves (extract)	What it looks like in the lab
Search group (2 moves)	Type-time function search; design search and extraction	Looking up literature methods; extracting an assay design from a standard
Generation group (3 moves)	Design patterns; logic generation; function generation	Applying control or block designs; generating SOPs and data tables
Thinking group (6 moves)	Problem solving, analysis, design, reasoning, thinking methods, R&D and exploration	Diagnosing failure, analysing data, causal reasoning, reverse validation
Evolution group (3 moves)	Continuous learning; refactoring and recombination; self-evolution	Learning across batches; refactoring the SOP; self-revision of the specification

1.3 Division of Labour in Mind AI: LLM Induces, Mind Model Deduces

The underlying platform is Mind AI. The LLM engine is good at language understanding, generation and the inductive statistics of the literature; the Mind Model engine is good at managing Type structure, deductive reasoning and long-term memory. Their division of labour in one line — the LLM speaks fast, the Mind Model speaks accurately; language goes to the LLM, intelligence goes to the Mind Model, and the ∩(T) criterion has the last word.

Fig. 1-3 Mind AI division of labour: the LLM handles induction and statistics, the Mind Model handles deductive reasoning

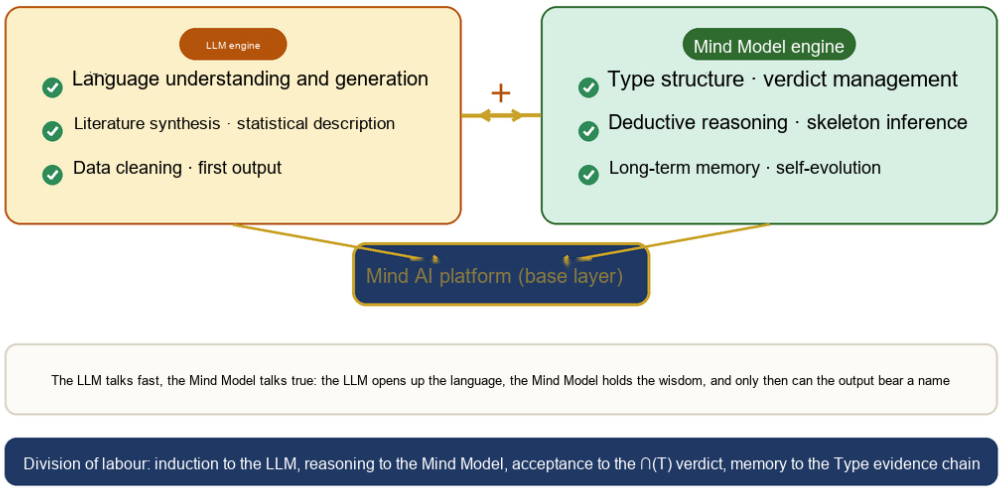


Figure 1-3 Division of Labour in Mind AI: the LLM handles induction and statistics, the Mind Model handles deduction

Table 1-4 Division of Labour: who is good at what, and at which station of the workflow

Engine	Strength	Workflow landing point
LLM engine	Language understanding and generation; induction and statistics	Literature review, first drafts, data cleaning, report polishing
Mind Model engine	Type structure, criteria, deductive reasoning, memory	Modelling the six Types, deriving the skeleton, ∩(T) acceptance, the evolution log
Mind AI platform	Integrates both engines plus reasoning, memory, learning and planning	Runtime environment and evidence-chain store for the S1–S7 pipeline

1.4 Preview: The Four Accuracy Gates

The phrase 'accurate output' resolves into four gates, climbed one at a time, each leaving checkable Type evidence behind. The next chapter shows what the outputs look like; Chapter 3 takes the gates apart one by one.

Table 1-5 Preview of the Four Accuracy Gates

Layer	Gate	Question	In the ALP case
Layer one	The ∩(T) criterion	Does the data pass?	Four gates: linearity R^2, CV, blank, dilution
Layer two	Reasoning skeleton	Is the theory pinned down?	v = Vmax[S]/(Km+[S])
Layer three	Experimental validation	Does the benchwork support it?	Controls, replication, statistical tests
Layer four	Type evidence chain	Does the conclusion trace back?	Conclusion → … → hypothesis: all six stations checkable

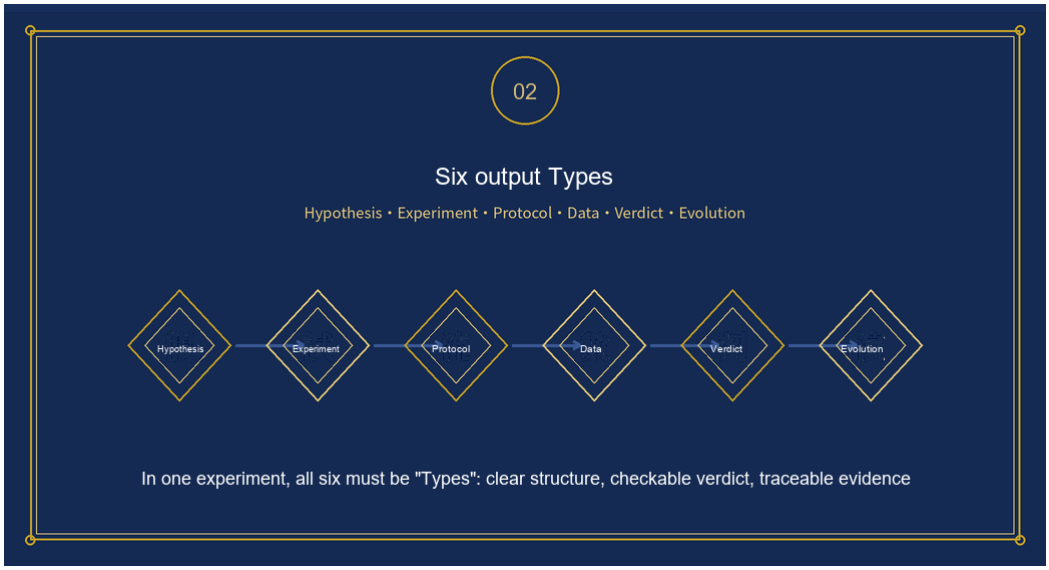
Chapter highlights

Experimental twin = Type + number: structure recorded as Type, measurement as number, and only then does the twin reason.

The two Types keep to their posts: Space Type governs what something is, Time Type governs how it is done.

Accuracy has a gatekeeper: the LLM induces and runs the statistics, the Mind Model deduces, and ∩(T) is the referee.

2 The Six Output Types: Hypothesis, Experiment, Protocol, Data, Verdict, Evolution



The output of the AI Bio-Lab Brain is not a pile of loose files but six interlocking Types. The six-word mnemonic runs hypothesis, experiment, protocol, data, verdict, evolution — Hypothesis Type, Experiment Type, Protocol Type, Data Type, Verdict Type, Evolution Type. Each Type has its fields, its criteria and its deliverables.

Figure 2-1 Six output Types at a glance: hypothesis, experiment, protocol, data, verdict, evolution — one click pushes the experiment to the end

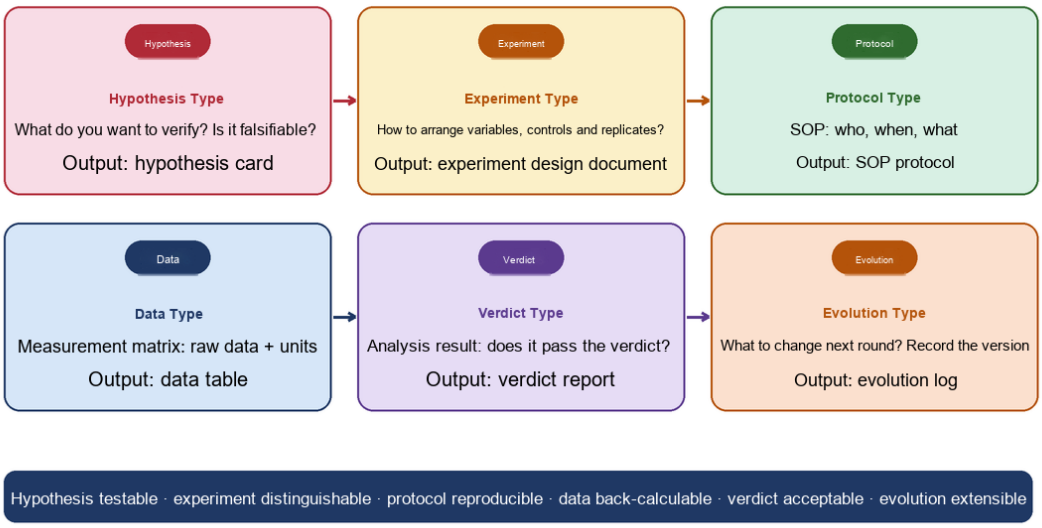


Figure 2-1 The Six Output Types at a Glance: one chain carries the experiment through

Table 2-1 The Six Types in Summary: each Type in one line

Output Type	English designation	Question answered	Deliverable	Acceptance criterion
Hypothesis Type (HypothesisType)	A falsifiable judgement statement	What do you want to verify?	Hypothesis card	Falsifiability conditions explicit
Experiment Type (ExperimentType)	Variables, controls, replication	How are variables and controls laid out?	Experiment design document	Independent / dependent / controlled clearly separated
Protocol Type (ProtocolType)	SOP action sequence	Who does what, and when?	SOP procedure	Reproducible by another operator
Data Type (DataType)	Measurement matrix and provenance	What was measured?	Data table plus raw values	All four ∩(T) gates passed
Verdict Type (VerdictType)	Parameters plus uncertainty	What is the conclusion?	Verdict report	Skeleton fits; residuals random
Evolution Type (EvolutionType)	Version narrative and lessons	What changes next round?	Evolution log	Every version has a story

Figure 2-2 Three brackets view one ALP kinetics assay: container, composition and abstract each tell their own story

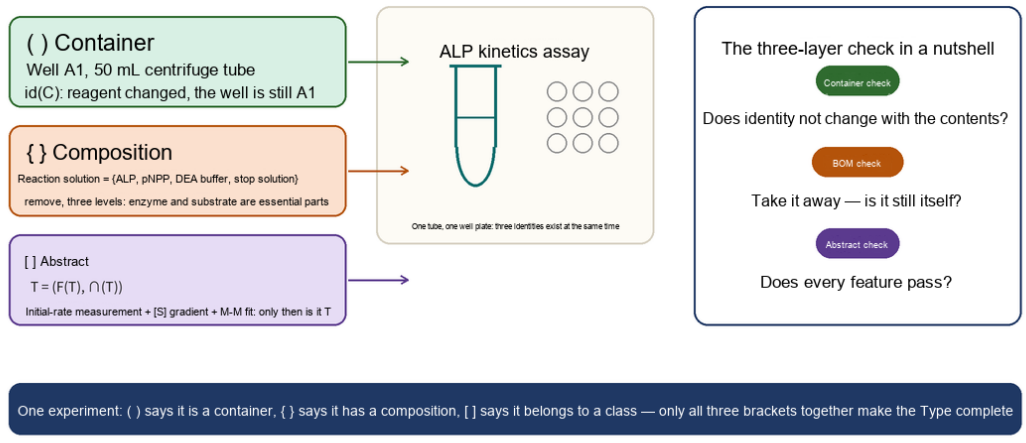


Figure 2-2 The Three Brackets Applied to an ALP Kinetics Assay: Container, Composition and Abstract each speak for themselves

2.1 Hypothesis Type (HypothesisType)

The Hypothesis Type is where the whole chain starts: what you want to verify, written as a falsifiable sentence. A vague aim ('get a feel for this batch of enzyme') is not a hypothesis; a statement that data can overturn ('the Km and Vmax of the new rALP batch fall within the

95% confidence interval of the previous batch') is. The Hypothesis Type also declares the variables and states the success criterion — every Type downstream draws on it.

Table 2-2 Hypothesis Type Field Card

Field	Required content	ALP case example
Hypothesis statement	One falsifiable judgement statement, with subject, variables and conditions all present	The new batch is kinetically equivalent to the previous batch
Alternative hypothesis	State what it looks like when it does not hold	Km or Vmax shifts materially
Falsifiability conditions	What kind of data would overturn it	Any parameter falling outside the previous batch's 95% CI
Variable declaration	Independent, dependent and controlled variables listed in one pass	See 2.2 Experiment Type
Success criterion	The numeric pass threshold, agreed before the work begins	Both parameters fall within the CI and all four gates pass

2.2 Experiment Type (ExperimentType)

The Experiment Type turns a hypothesis into a design that can discriminate: which independent variable is moved, which dependent variable is read, which controlled variables are locked down, how many control groups, how many replications. A poorly designed experiment cannot be rescued by refined data later on — so acceptance of the Experiment Type asks one question only: can this design tell the two worlds apart?

Figure 2-3 The skeleton of the Type: independent, dependent and control variables distinguished at once

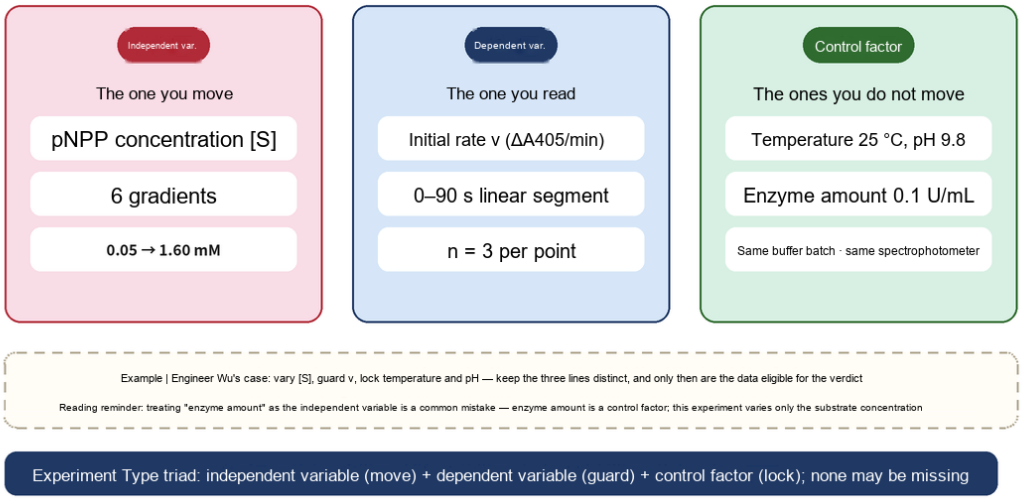


Figure 2-3 The backbone of the Experiment Type: independent, dependent and controlled variables separated at a glance

Table 2-3 The three-way variable design card (ALP kinetics case)

Variable	Role	Setting in this case	Common error
pNPP concentration	Independent variable	6 levels, 0.05 → 1.60 mM	The gradient does not bracket Km, so the fit is unstable
Initial rate v	Dependent variable	ΔA405/min, linear segment over 0–90 s	Reading the endpoint value and passing it off as the initial rate
Temperature / pH / enzyme amount	Controlled variable	Locked at 25 °C, pH 9.8, 0.1 U/mL	Mistaking the enzyme amount for an independent variable
Control group	Required provision	Blank + heat-denatured enzyme + previous-batch standard	Skipping controls; it cannot be made up afterwards
Replication	Required provision	n = 3 per point, randomised run order	Drawing a conclusion at n = 1

The Experiment Type also conceals the remove criterion of the Space Type: a reaction recipe is a bill of materials, and before removing any single item you must ask, "once it is removed, is the whole still the same reaction?" The answer decides whether the item is an essential component, a functionally degraded component, or a non-essential component.

Fig. 2-4 The remove criterion applies to reaction composition: take one part away and see what remains of the whole

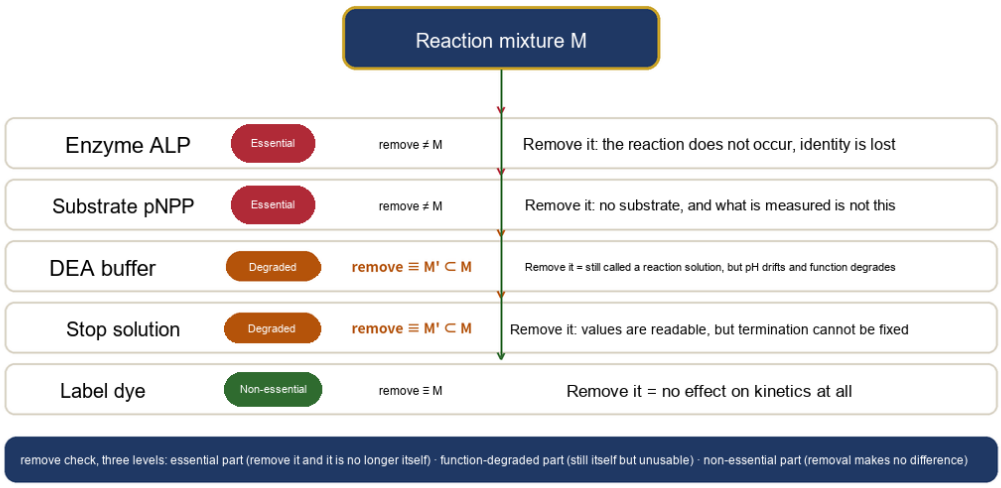


Figure 2-4 The remove criterion applied to reaction composition: take one item away and see what is left of the whole

Table 2-4 The three remove grades × reaction composition

Composition	Grade	remove criterion	Rationale
Enzyme (ALP)	Essential component	remove(p,M) ≢ M	No enzyme, no reaction: identity is lost
Substrate (pNPP)	Essential component	remove(p,M) ≢ M	Without substrate, the assay measures something else
DEA buffer	Functionally degraded component	remove ≡ M' ⊂ M	pH drifts; still called the reaction mixture, but no longer usable
Stop solution	Functionally degraded component	remove ≡ M' ⊂ M	No fixed-point termination, so readings lose accuracy
Labelling dye	Non-essential component	remove(p,M) ≡ M	Unrelated to kinetics; removing it makes no difference

2.3 Protocol Type (ProtocolType)

The Protocol Type is the Type of the SOP: it turns the experimental design into a sequence of actions — who, when, does what, to what degree. Every action maps onto a move of the Time Type (pipetting embodies function generation, checking instrument calibration is search, exception handling is problem solving). The Protocol Type has a single acceptance standard: hand it to someone else, and they obtain data of the same grade.

Table 2-5 Protocol Type (SOP) field card

Field	Required content	Illustrative example
Protocol number	Unique code + revision + effective date	SOP-ALP-042 v2.1
Scope	Which Experiment Type and Hypothesis Type it corresponds to	Inter-batch kinetics comparison for rALP
Action sequence	Steps, parameters, tolerance, corresponding move	Pre-heat 5 min at 37 °C → start by adding pNPP
Key control points	Which steps drift most easily, and how to prevent it	Reaction start timing ±2 s, temperature control ±0.5 °C
Exception handling	Branches and recording method when something goes wrong	Blank drift over the limit → re-make the buffer

2.4 Data Type (DataType)

The Data Type is the Type of measurement: raw value, unit, instrument, timestamp, operator — not one field may be omitted. It carries two disciplines. First, raw values are never overwritten; cleaning is saved as a separate version. Second, before reaching the verdict, data must pass the ∩(T) four gates (set out in Chapter 3) — data that fails the gates is not "poorer data", it is "data that may not be used".

Table 2-6 Data Type field card

Field	Required content	Illustrative example
Measurement matrix	Raw values by sample × condition × replication	6 concentrations × 3 replicates = 18 initial-rate values
Units and range	SI units, instrument range and resolution	ΔA405/min, range 0–3 A
Traceability triad	Instrument ID, calibration record, operator	UV-2600#0412, CAL-2603, Engineer Wu
Timestamp	Measurement time and batch sequence	2026-10-14 09:20–10:05
Version discipline	Raw values read-only; cleaning saved as a new v with differences logged	raw v1 → clean v1 (blank subtracted)

2.5 Verdict Type (VerdictType)

The Verdict Type turns data into a conclusion, but the conclusion must grow on the reasoning skeleton: recognise the equation first (Michaelis–Menten), then let the data

support or overturn it. Its output is not a lone number but a set — parameter + uncertainty + applicable range + comparison verdict. Stating the range explicitly (for example, "valid only within 0.05–1.60 mM") is what makes a verdict responsible.

Table 2-7 Verdict Type field card

Field	Required content	Illustrative example
Skeleton and method	Which equation, which fit	Primarily M–M non-linear regression
Parameters and uncertainty	Point estimate ± SD/CI, not just the mean	Km = 0.25 ± 0.02 mM
Residual check	Residual randomness; treatment of outliers	Residuals random, no systematic offset
Scope	Boundary conditions within which the conclusion is valid	[S] 0.05–1.60 mM, 25 °C
Comparison verdict	Checked against the hypothesis: pass / fail	Falls within the previous batch's 95% CI → equivalence holds

2.6 Evolution Type (EvolutionType)

The Evolution Type lets the lab's knowledge grow: this round's conclusions and lessons are written into the evolution log and become the nutrient for the next round's hypothesis and the reason for revising the SOP. Its acceptance standard is "every version has a story" — each version of the SOP can answer: what changed, why it changed, what will be tested next time. A lab without an Evolution Type can run a thousand experiments in ten years and in truth has run one experiment a thousand times.

Table 2-8 Evolution Type field card

Field	Required content	Illustrative example
Version narrative	What changed and why	v2.1→v2.2: added the temperature variable
Lesson log	Pitfalls encountered and how they were solved	Blank drift → buffer made fresh the same day
Next-round hypothesis	New questions grown from this round's conclusions	Temperature dependence of Km at 25/30/37 °C
Trigger criterion	What condition starts a new round	Triggered when two consecutive batches fall outside the CI

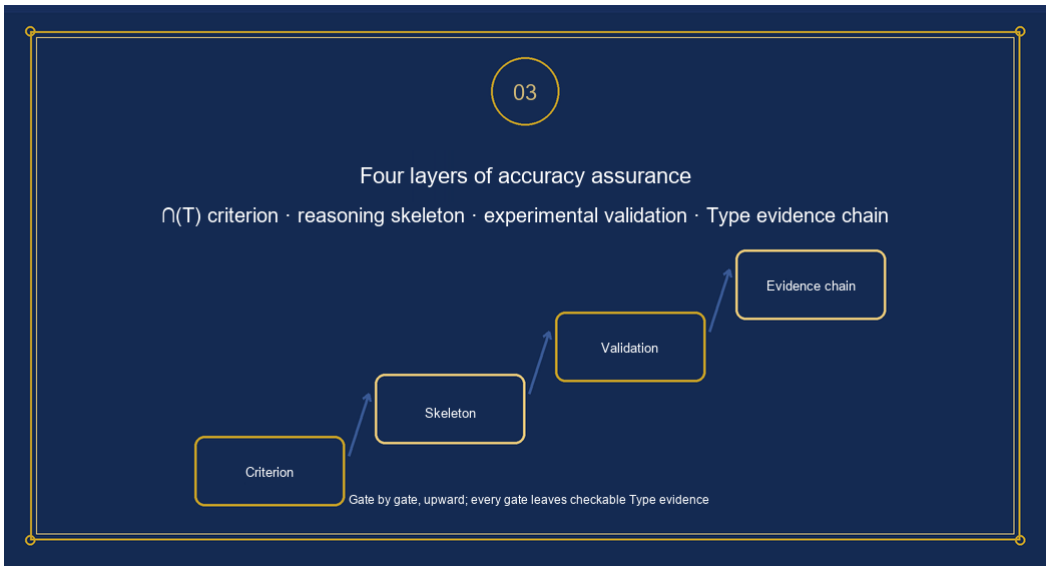
Chapter highlights

The six-Type mnemonic "hypothesis, experiment, protocol, data, verdict, evolution": the hypothesis is falsifiable, the experiment discriminable, the protocol reproducible, the data recomputable, the verdict acceptable, the evolution continuable.

The three bracket types and the remove criterion are hidden inside the experiment: container identity, recipe composition and category criterion should all be settled when the project opens.

Every Type has its field card and acceptance criteria — output does not rest on a feeling, it rests on the Type.

3 The Four Layers of Accuracy Assurance



"Accurate output" cannot be guaranteed by intuition; it needs gates. The four layers run bottom-up: the ∩(T) criterion keeps data outside or lets it in, the reasoning skeleton pins the theory down, experimental validation lets the bench have its say, and the Type evidence chain lets a conclusion be traced back to its source; finally the Space × Time cross-acceptance keeps overall watch.

3.1 Layer One | The ∩(T) Criterion: Four Data-Quality Gates

An abstract category T is determined by its feature set F(T) and the intersection criterion ∩(T): an item belongs only if it passes every feature. Put that ruler on data and you get the four data-quality gates — linear region, repeatability, blank, dilution linearity. Only when all four pass is the Data Type qualified; if any one fails, the data goes back for re-measurement, and "fit first, argue later" is not permitted.

Fig. 3-1 First layer | ∩(T) Data-quality four-gate funnel: data must clear every gate before it may be aggregated



Figure 3-1 The ∩(T) four-gate data-quality funnel: data must pass the gates before it may enter the fit

Table 3-1 The ∩(T) four-gate criterion card (ALP case)

Gate	Criterion (agreed up front)	Measurement method	If the gate fails
Gate 1 · Linear region	Initial rate linear over 0–90 s, $R^2 \geq 0.99$	Plot A405 against time and take the slope	Reduce the enzyme amount and re-measure
Gate 2 · Repeatability	Three replicates at the same point, CV ≤ 5%	Within-group CV of the 18 values	Check pipetting and timing error
Gate 3 · Blank	Blank drift ≤ 0.01 ΔA/min	Read the no-enzyme control throughout	Re-make the buffer and substrate
Gate 4 · Dilution linearity	Back-calculated $R^2 \geq 0.99$ at 0.5×/1×/2×	Back-calculate activity from three enzyme concentrations	Check for enzyme inactivation or substrate inhibition

Worked example | How the four gates are recorded

Gate 1: $R^2 = 0.997$ (0–90 s, six time points) Gate 2: CV = 1.8% (calculated in groups across the 18 values) Gate 3: blank drift = 0.004 ΔA/min Gate 4: dilution back-calculation $R^2 = 0.999$ → all four gates pass; the Data Type is certified "qualified".

3.2 Layer Two | The Reasoning Skeleton: Michaelis–Menten

The reasoning skeleton means: recognise the equation first, then let the data support or overturn it — not "draw a curve that looks agreeable". The skeleton of enzyme kinetics is the Michaelis–Menten equation $v = V_{max}\cdot[S]/(K_m+[S])$; it also supplies the derivation chain: the double-reciprocal form $1/v = (K_m/V_{max})\cdot(1/[S]) + 1/V_{max}$ gives slope and intercept their separate meanings, and the half-maximal point gives Km directly. With the skeleton

standing first, the fit has a referee.

Figure 3-2 Layer 2 | reasoning skeleton: the Michaelis-Menten equation grounds the data in theory

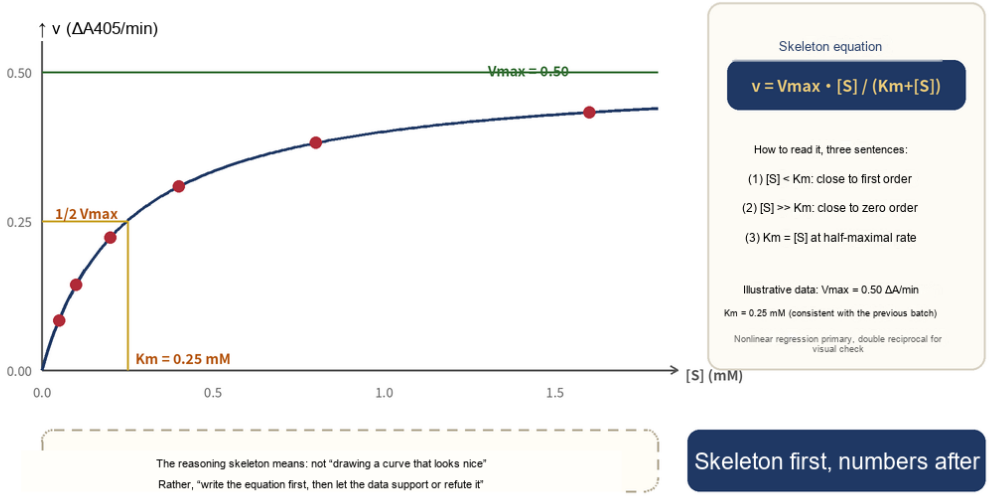


Figure 3-2 The reasoning skeleton: the equation pins the data to the theory

Figure 3-3 Double-reciprocal check: the Lineweaver-Burk plot straightens the curve for acceptance

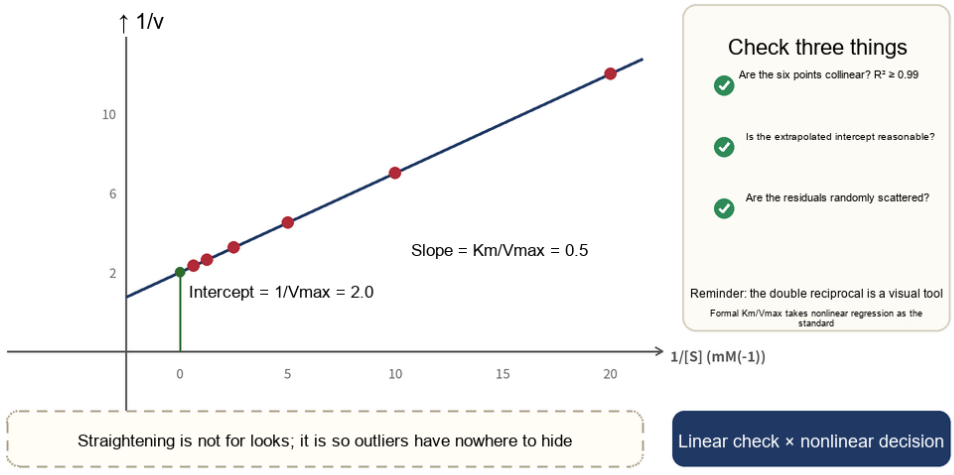


Figure 3-3 The double-reciprocal check: straightening the curve for acceptance

Table 3-2 Reasoning-skeleton derivation steps (ALP case)

Step	Action	Purpose and output
1	Write the skeleton: v = Vmax[S]/(Km+[S])	Theory first; two parameters to estimate, Vmax and Km
2	Derive the double reciprocal: 1/v = (Km/Vmax)(1/[S]) + 1/Vmax	Slope = Km/Vmax, intercept = 1/Vmax, each with physical meaning
3	Non-linear regression for Vmax, Km and the 95% CI	The formal result rests on the non-linear fit; the double reciprocal is for inspection only

Step	Action	Purpose and output
4	Residual check + treatment of outliers	Accept only random residuals; investigate outliers experimentally before discussing statistics
5	Compare with the previous batch's specification (CI overlap test)	Produces the comparison verdict of the Verdict Type

3.3 Layer Three | Experimental Validation: Control, Replication, Statistics

However elegant the skeleton, the bench must nod in agreement. The third layer asks three plain things: are the controls complete, is the replication sufficient, does the statistics hold up. Control is an essential component of the Experiment Type (the remove criterion: take the control away and the experiment loses its identity); replication supplies the uncertainty; statistics translates "about the same" into a confidence interval.

Table 3-3 The experimental-validation triad

Item	Configuration in this case	Question it answers
Control	Blank + heat-denatured enzyme + previous-batch standard	Does the signal come from the reaction? Is the enzyme activity real? Is it comparable?
Replication	n = 3 per point, independently prepared solutions, randomised run order	How large is the error? Is the CV within 5%?
Statistics	Non-linear regression + 95% CI + residual inspection	Are the parameters estimated accurately? Is the difference from the previous batch significant?

3.4 Layer Four | The Type Evidence Chain

The last layer is responsible for tracing back: any conclusion can be followed along Verdict Type, Data Type, Protocol Type, Experiment Type and Hypothesis Type all the way to its source. Break one link and the conclusion is demoted to an opinion for reference; only a complete chain may be written into a report, and only then is it fit to be signed.

Figure 3-4 Layer 4: Type evidence chain — trace the conclusion backwards, every station has a checkable Type

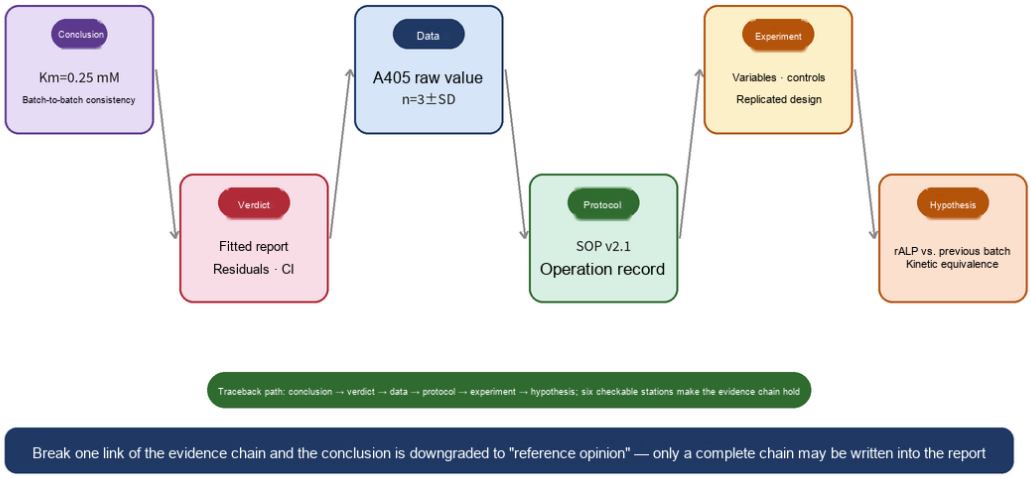


Figure 3-4 The Type evidence chain: trace the conclusion backwards, and every station has a Type to check

Table 3-4 The six stations of the Type evidence chain (ALP case)

Station	What is checked	Evidence in the ALP case
Conclusion	The conclusion sentence and its applicable range	Km = 0.25 mM, range 0.05–1.60 mM
Verdict Type	Fit method, residuals, CI	Non-linear regression report v1
Data Type	Raw values and the four-gate record	18 initial-rate values + sign-off that all four gates passed
Protocol Type	SOP version and run record	SOP-ALP-042 v2.1, item-by-item trace
Experiment Type	Variable, control and replication design	Experiment design document EX-2610-07
Hypothesis Type	Falsifiability conditions and success criterion	Hypothesis card H-042: equivalence criterion established first

3.5 Space × Time Cross-Acceptance

The four layers in fact have two faces: the Space Type checks whether the structure is right, the Time Type checks whether the protocol will run. The two faces cross into four cells — only the cell where Space is right and Time is also right is cleared for reporting; the other three each have a clear remedial path.

Figure 3-5 Space × time cross-acceptance: only when the structure holds and the process runs does it count as certain

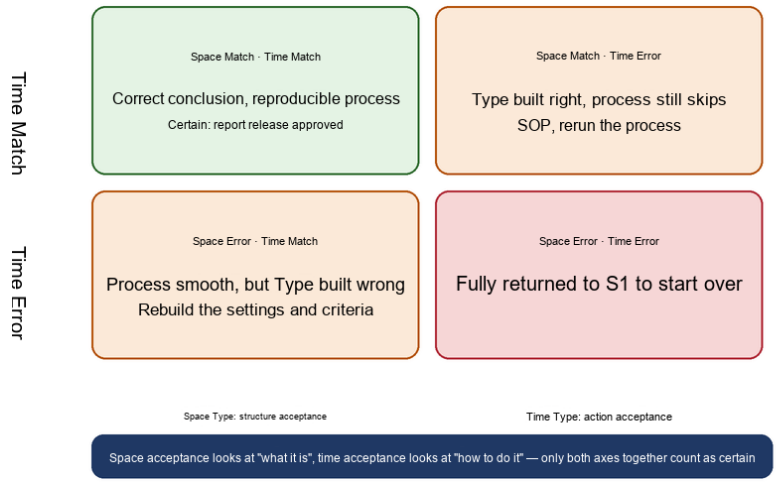


Figure 3-5 The Space × Time cross-acceptance matrix

Table 3-5 Cross-acceptance table

Cell	Symptom	Remedy
Space right · Time right	Structure correct, protocol reproducible	Cleared for reporting; entered into the Type evidence chain
Space right · Time wrong	The Type is built correctly but the protocol jumps around	Top up SOP training, then re-run the protocol segment
Space wrong · Time right	The protocol runs smoothly but the Type is built crooked	Go back to step 2, build the Type along the axes, and rebuild the definition and criterion
Space wrong · Time wrong	Total breakdown	Return to S1 and start again; seal the old outputs as lessons

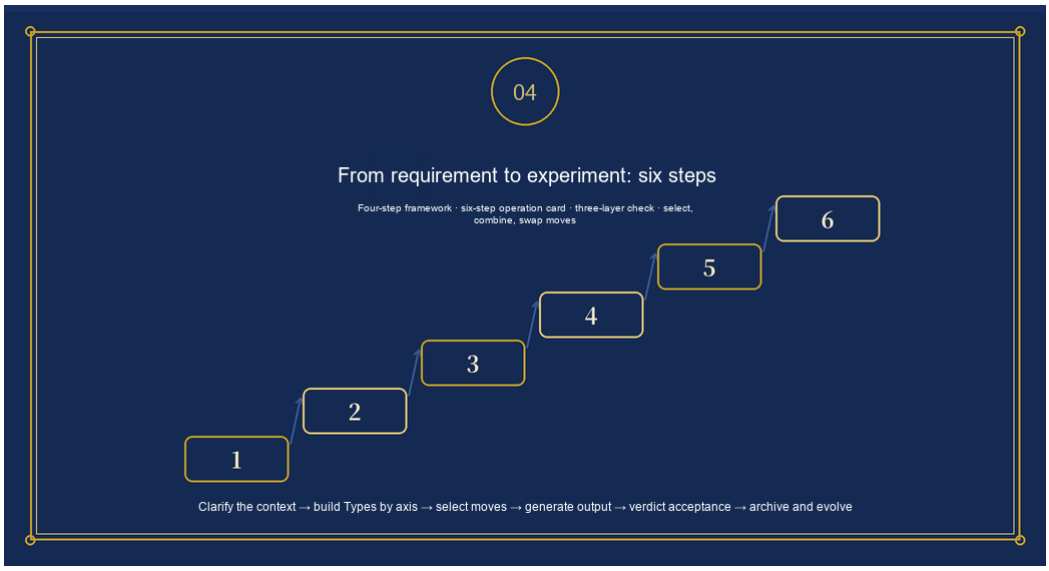
Chapter highlights

Four layers, none to be skipped: the ∩(T) criterion → the reasoning skeleton → experimental validation → the Type evidence chain.

Skeleton first, data second: recognise the equation first, then let the data speak.

Cross-acceptance: Space checks the structure, Time checks the protocol; only when both axes pass is the result accurate.

4 From Requirement to Experiment: The Six Steps



Having met the six Output Types and the four layers, this chapter arranges them into a single operating path: the four-step framing pins the requirement down first, the six steps run from context all the way to evolution, and along the way a three-layer judgement keeps watch while the picking, combining and swapping of moves dispatches the Time Type's moves.

4.1 The Four-Step Framing: Recognising the Requirement

Establishing a requirement is not a matter of deciding in a meeting off the top of one's head; it is a small exercise in abstraction engineering: distinguish similarities and differences → trigger abstraction → name the dimension → assign membership. When a new set of differences refuses to match the existing specification (a failed control), that is the moment a new dimension is forced out — once the dimension has a name, the requirement can be assigned back to one of the six Output Types, and the remaining five steps have a target.

Figure 4-1 Four-step framework to clarify requirements: a failed control forces a new dimension; once the dimension is fixed, the requirement becomes clear

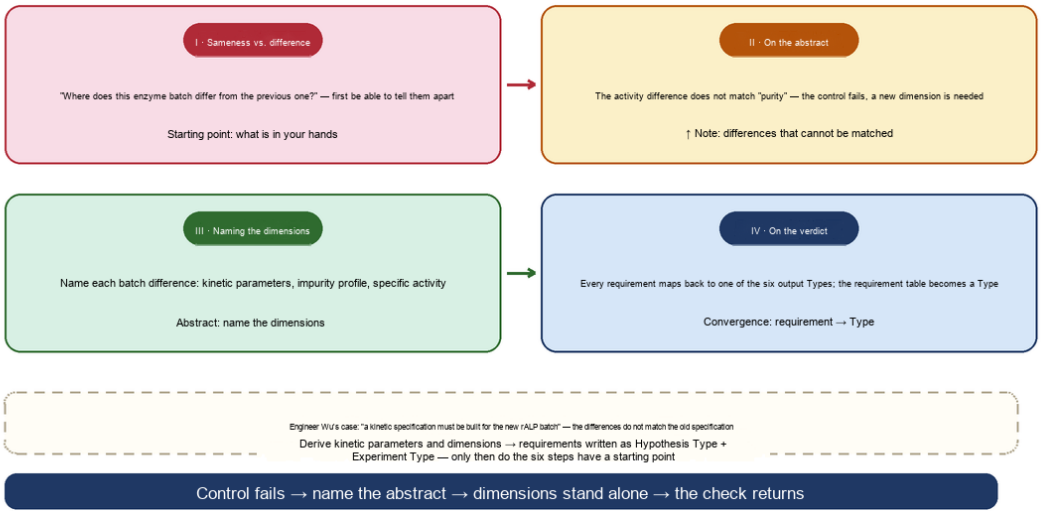


Figure 4-1 The four-step framing recognises the requirement: a failed control forces a new dimension

Table 4-1 The four-step framing × opening an experiment

Step	What you do	In the ALP case
I · Distinguish similarities and differences	First be able to say where this batch differs from the previous one	The new batch's activity readings do not match the old specification
II · Trigger abstraction	When the difference does not fit the old dimensions, a new dimension is needed	"Purity" cannot explain the difference → it is a kinetic parameter
III · Name the dimension	Name the difference and the dimension stands on its own	Three new dimensions: Km, Vmax and specific activity
IV · Assign membership	Assign the requirement back to one of the six Types and write it as a Type entry	→ one Hypothesis Type entry and one Experiment Type entry

4.2 The Six-Step Operation Card

The six steps form a loop: identify the context → build the Type along the axes → select and expand → generate the output → acceptance against the criterion → archive and evolve. The evolution log from step 6 is the context input to step 1 of the next round — the loop has no end, and the lab's knowledge grows inside it.

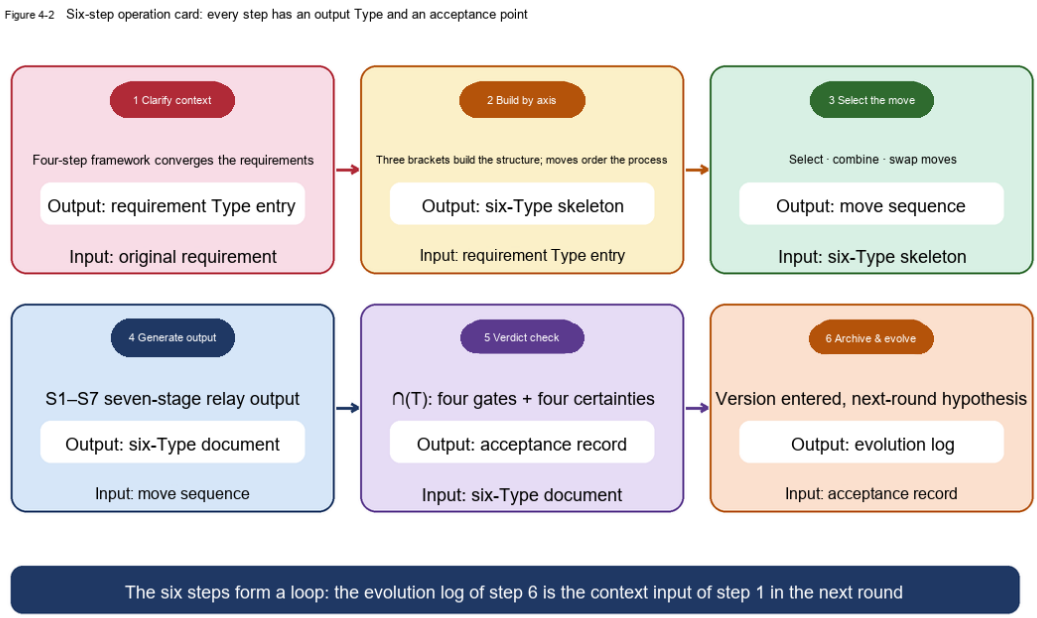


Figure 4-2 The six-step operation card: every step has an Output Type and an acceptance point

Table 4-2 The six-step operation card: action, Output Type, moves, acceptance

Step	Action	Output Type	Commonly used moves	Acceptance point
1 Identify the situation	The four-step framing converges the requirement	Requirement Type entry	Problem solving, analysis	Requirement falsifiable
2 Split the axes, build the Types	The three bracket types build the structure; the moves order the protocol	Six-Type skeleton	Design, design patterns	Skeleton complete with all six Types
3 Select the expansion pattern	Selecting moves, combining moves, switching moves	Move sequence	(whole-board deployment)	Sequence executable
4 Generate the outputs	S1–S7 seven-stage relay production	Six-Type documents	Function generation, logic generation	Each Type passes its field card
5 Criterion-based acceptance	The ∩(T) four gates plus four-layer accuracy	Acceptance record	Reasoning, analysis	All four gates passed
6 Archive and evolve	Version logged, next-round hypothesis	Evolution log	Continuous learning, self-evolution	Every version has a story

4.3 Three-Layer Judgement: Selecting, Combining and Switching Moves

Two standing tools are kept on hand along the operating path. The first is the three-layer judgement (is / has / matches): the Container judgement asks 'is it the same one', the BOM judgement asks 'is anything missing', and the Abstract judgement asks 'is it the right class'

— any qualification check is swept with these three questions. The second is move deployment: selecting a move (one move for a single situation), combining moves (several moves in relay), switching moves (change the route when stuck) — switching moves never switches the criteria; only the route changes, and not one ∩(T) gate may be dropped.

Figure 4-3 The three-layer check applied to experiments: the container asks identity, the BOM asks composition, the abstract asks classification

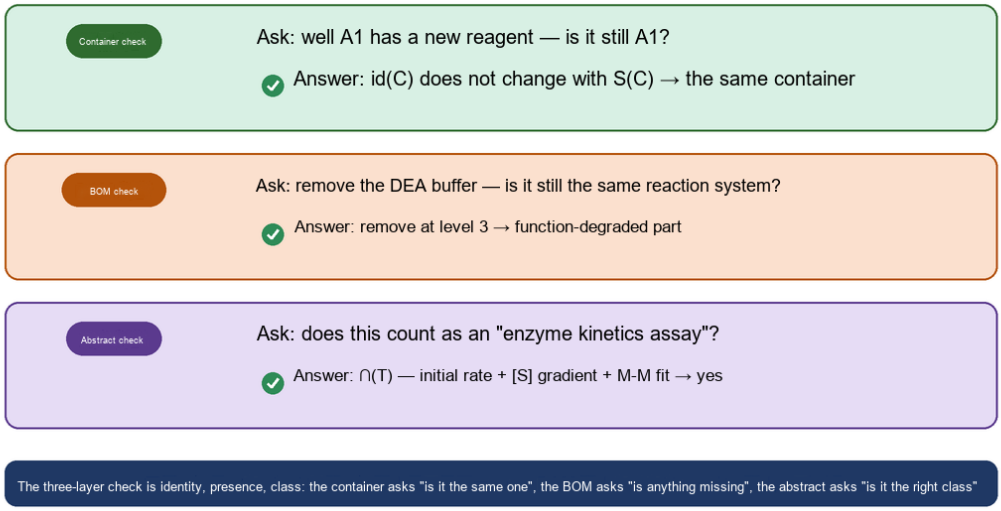


Figure 4-3 Three-layer judgement: Container asks identity, BOM asks Composition, Abstract asks classification

Figure 4-4 Select, combine, swap: the Time Type's move palette, deployed to context

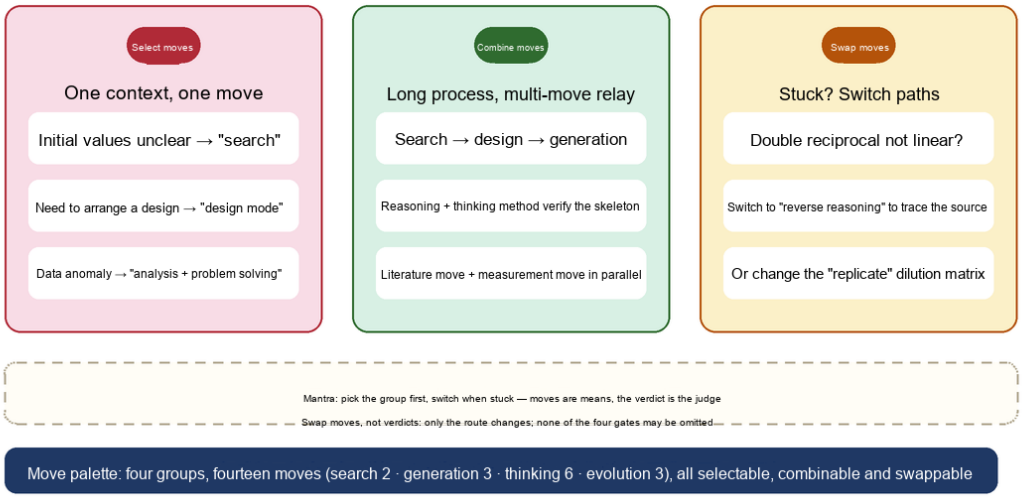


Figure 4-4 Selecting, combining and switching moves: the move board is deployed to suit the situation

Table 4-3 Three-layer judgement scan table (ALP case)

Layer	Question	Answer in the ALP case
Container judgement	Does identity survive a change of contents?	Well A1 remains A1 after a reagent change → the same Container

Layer	Question	Answer in the ALP case
BOM judgement	Does it still recognise itself once taken apart?	Removing the buffer would make it a function-degraded part, so it is not stripped
Abstract judgement	Does it pass on every feature?	Initial rate + [S] gradient + M–M fit all pass → belongs to this class

Table 4-4 Selecting, combining and switching moves compared

Move	When to use it	Illustrative example
Selecting a move	One move for a single situation	Unclear initial values → Search; a design to be laid out → design patterns
Combining moves	Several moves in relay over a long protocol	Search → Design → Function generation; Reasoning + Thinking to check the skeleton
Switching moves	Change the route when stuck	Double reciprocal not linear → reverse reasoning to trace the source, rebuild the dilution matrix

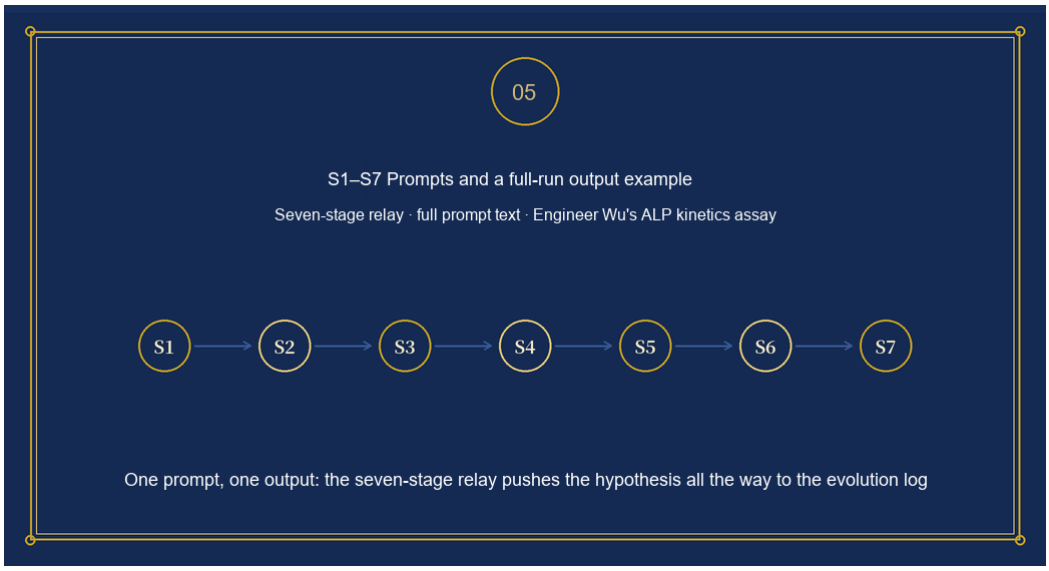
Chapter highlights

Identify the requirement before acting: the four-step framing turns 'the control failed' into a new dimension, and only then can the requirement be written into shape.

The six steps form a loop: the evolution log feeds back into the next round of situations.

The three-layer judgement (is / has / matches) is a standing scan; moves may be selected, combined or switched, but the criteria never change.

5 S1–S7 Prompts and a Full-Run Output Example



This chapter is the operational core of the book: the full prompt text of the S1–S7 seven-stage relay is handed to you first, and then Engineer Wu's ALP kinetics assay runs the outputs produced by all seven stages from beginning to end. The prompts can be copied straight out and used once modified; every example datum is marked illustrative.

5.1 The Seven-Stage Relay

The rhythm of the seven-stage relay is: define clearly → design to discriminate → protocol reproducible → data recomputable → verdict acceptable → report traceable → evolution continuable. Each stage consumes the output of the previous stage and emits the input of the next; the evolution log from S7 feeds back into S1, and the next round of hypotheses grows from there. There are six rules of use, and keeping them beside the prompts is the safest arrangement.

Figure 5-1 S1–S7 seven-stage relay: each stage consumes the previous output and emits the next input

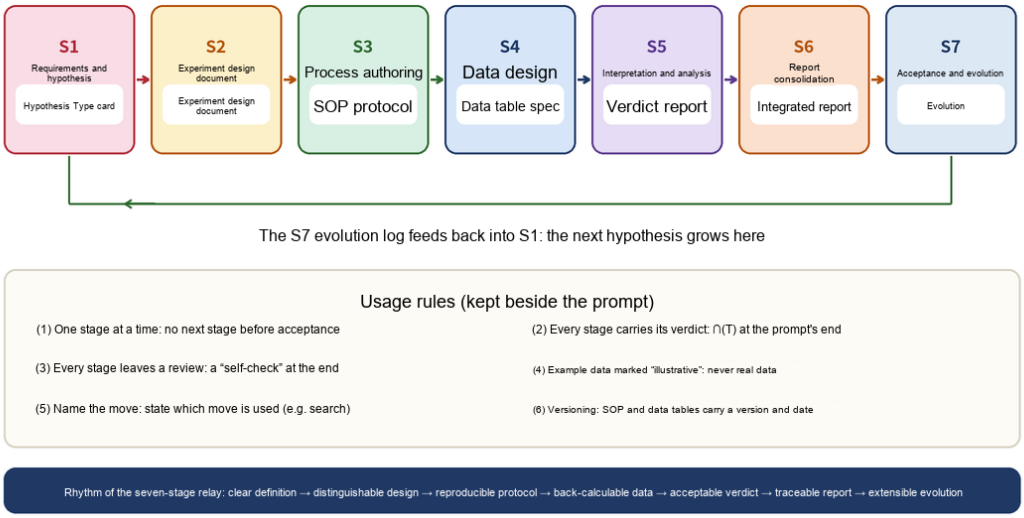


Figure 5-1 The S1–S7 seven-stage relay: each stage consumes the previous stage's output and emits the next stage's input

Table 5-1 S1–S7 overview

Stage	Task	Output Type	Commonly used moves	Stage-pass criterion
S1	Requirement and hypothesis	Hypothesis Type card	Problem solving, Thinking	Falsifiability conditions explicit
S2	Experiment design	Experiment design document	Design, design patterns	Three-way variable split clear; controls complete
S3	Protocol drafting	SOP procedure	Function generation, logic generation	Reproducible by another operator
S4	Data design	Data table specification	Logic generation	Four-gate criteria built in
S5	Verdict analysis	Verdict report	Analysis, reasoning	Skeleton fits; residuals random
S6	Report consolidation	Integrated report	Logic generation	Evidence chain complete
S7	Acceptance and evolution	Evolution log	Continuous learning, self-evolution	Every version has a story

5.2 Full Prompt Text for S1–S7

The seven prompts below are the substance of the 'method for using outputs accurately' that this book delivers. Every stage follows the same format: task, input, output, criteria, and the three self-check questions. Open only one stage at a time; if the previous stage's output

has not been accepted, do not open the next.

S1 prompt | Requirement and hypothesis

You are an [experiment design analyst]. Task: converge a vague experimental requirement into a falsifiable Hypothesis Type card (HypothesisType).
Input: requirement description, background material, (optional) previous batch specification or literature.
Output: a hypothesis card with five fields — hypothesis statement (one falsifiable declarative sentence), opposing hypothesis, falsifiability conditions, variable declaration (independent / dependent / controlled), success criterion (numerical threshold).
Criteria: the statement must contain a subject, a variable and a condition; the success criterion must be a number or a decidable statement; points of uncertainty must be listed explicitly in a 'to be confirmed' list.
Three self-check questions: (1) Can this statement be overturned by data? (2) Was the success criterion stated in advance? (3) Is the three-way variable declaration complete?

S2 prompt | Experiment design

You are an [experiment design engineer]. Task: translate the Hypothesis Type card into an Experiment Type design document (ExperimentType) that can tell the two worlds apart.
Input: Hypothesis Type card, list of available instruments and materials, time and sample budget.
Output: experiment design document — independent variable (gradient and range), dependent variable (measurement and unit), controlled variables (locked values), control groups (blank / positive / reference standard), replication and randomisation, three-level remove analysis (recipe BOM breakdown).
Criteria: the independent / dependent / controlled split is mutually exclusive and complete; the controls can tell 'where the signal comes from'; the number of replications matches the budget; the gradient spans the expected transition point (e.g. around Km).
Three self-check questions: (1) Can this design discriminate the two worlds? (2) Is any control missing? (3) Does the gradient cross the transition point?

S3 prompt | Protocol drafting (SOP)

You are a [protocol engineer]. Task: translate the experiment design document into a Protocol Type SOP (ProtocolType) that another operator can reproduce, annotating the corresponding move at every step.
Input: experiment design document, instrument operating limits, safety and regulatory considerations.
Output: SOP — protocol number + revision + effective date, scope of application, action sequence (step / parameter / tolerance / move), key control points, exception-handling branches.
Criteria: every step has a tolerance and a timing origin; key control points state 'where it is most likely to drift'; exception handling has branches and requires a record.
Three self-check questions: (1) If someone else follows it, will they obtain data of the same grade? (2) Is a tolerance written for every step? (3) Is it clear what to do when something goes wrong?

S4 prompt | Data design and measurement

You are a [data engineer]. Task: translate the SOP into Data Type (DataType) measurement specifications and a record table, building the ∩(T) four gates directly into the table.
Input: SOP, instrument range and resolution, (optional) previous batch data format.
Output: data table specification — measurement matrix (sample × condition × replication), units and range, the three traceability items (instrument / calibration / operator), timestamp fields, version discipline (raw values read-only), four-gate criterion columns (linearity / replication / blank / dilution).
Criteria: every measured value traces back to an instrument and a person; raw and cleaned values are stored separately; the four-gate criterion columns are built in and pre-filled with the 'planned threshold'.
Three self-check questions: (1) Can the raw value be overwritten (it must not be)? (2) Are units and range complete? (3) Are the four-gate columns in place?

S5 prompt | Verdict analysis

You are a [statistical verdict engineer]. Task: produce a Verdict Type report (VerdictType) from data that has already passed the four gates, with the conclusions growing on the reasoning skeleton.
Input: qualified Data Type, Hypothesis Type card, skeleton equation (e.g. Michaelis–Menten).
Output: verdict report — skeleton and fitting method, parameters ± uncertainty (CI), residuals and outlier handling, range of applicability, comparison verdict against the hypothesis / previous batch (pass / fail).
Criteria: recognise the equation before fitting (skeleton first); a point estimate reported without a CI counts as incomplete; the range of applicability must state its boundaries; correlation must not masquerade as causation.
Three self-check questions: (1) Does the conclusion exceed the measured range? (2) Have the residuals been examined? (3) Has the uncertainty been reported?

S6 prompt | Report consolidation

You are the [lead author of the experimental report]. Task: consolidate the outputs of the six Types into an integrated report and attach the complete Type evidence chain.
Input: Hypothesis Type card, experiment design document, SOP, Data Type, Verdict Type report.
Output: integrated report — abstract, one section for each of the six Types, evidence-chain index (conclusion → verdict → data → protocol → experiment → hypothesis, six linked stations), limitations and declarations.
Criteria: every concluding sentence carries an evidence-chain index; abstract figures agree with the body; illustrative and real data are marked in clearly separated zones.
Three self-check questions: (1) Can every conclusion be traced back? (2) Do the abstract and body figures agree? (3) Are the limitations stated honestly enough?

S7 prompt | Acceptance and evolution

You are the [laboratory quality officer]. Task: perform four-layer accuracy acceptance on the integrated report and write the conclusions and lessons into the Evolution Type log (EvolutionType).
Input: integrated report, four-layer criteria checklist, (optional) previous evolution log.
Output: acceptance record (four gates + cell-by-cell sign-off of the cross-acceptance) + evolution log (version narrative / lesson register / next-round hypothesis / trigger criterion).
Criteria: each of the four layers is signed off individually — never 'probably passed'; the three evolution-log fields (what changed / why it changed / what to check next time) are complete; failed items list a remedy path explicitly.
Three self-check questions: (1) Were the four gates actually signed? (2) Does the version have a story? (3) Has the next-round hypothesis grown out?

5.3 Engineer Wu's ALP Kinetics Assay

Situation: Engineer Wu at a biotech R&D laboratory receives a new batch of recombinant alkaline phosphatase (rALP, lot ALP-2610). Under the quality specification, the kinetic parameters of the new batch must be equivalent to those of the previous batch (ALP-2604). Engineer Wu decides to run a complete seven-stage relay in order to establish the kinetic specification for the new batch. All outputs below are illustrative.

Figure 5-2 Engineer Wu's case · six Type cards issued throughout: every Type has a deliverable and an acceptance

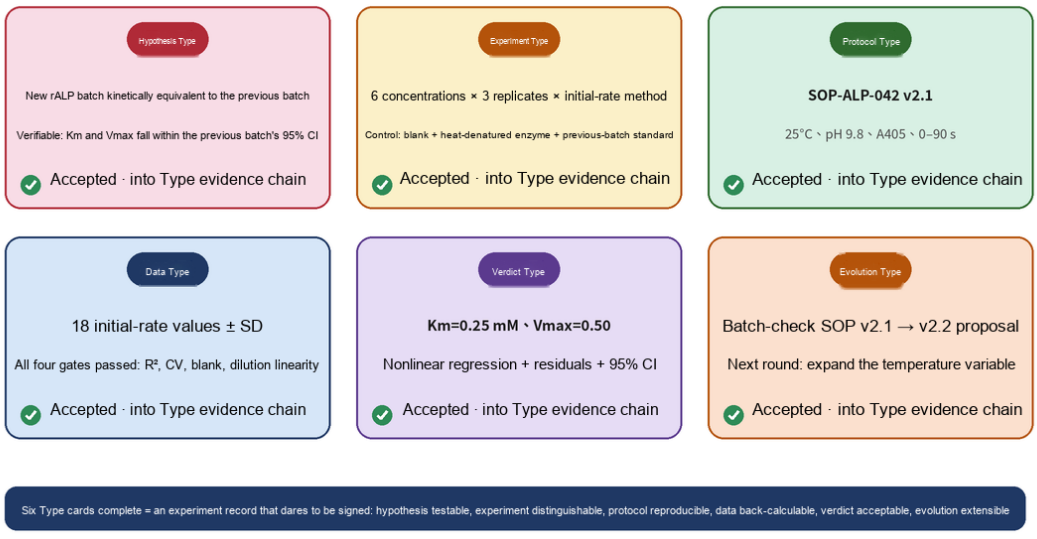


Figure 5-2 The six Type cards produced across the run: every Type has a deliverable and an acceptance check

Table 5-2 S1 output · Hypothesis Type card H-042 (illustrative)

Field	Content
Hypothesis statement	The Km and Vmax of the new rALP batch ALP-2610 fall within the 95% confidence intervals of the previous batch ALP-2604 (kinetic equivalence)

Field	Content
Alternative hypothesis	Either parameter falls outside the previous batch's 95% CI, or a different mechanistic signature appears (a change in the double-reciprocal plot pattern)
Falsifiability conditions	After non-linear regression, the hypothesis is overturned if the CI of Km or of Vmax does not overlap the previous batch's CI
Variable declaration	Independent variable: pNPP concentration; dependent variable: initial rate v (ΔA405/min); controlled variables: 25 °C, pH 9.8, enzyme loading 0.1 U/mL
Success criterion	Both parameters fall within the previous batch's 95% CI, and all four ∩(T) gates pass (linearity $R^2 \geq 0.99$, CV ≤ 5%, blank ≤ 0.01, dilution $R^2 \geq 0.99$)

Table 5-3 S2 output · Experiment design document EX-2610-07 (illustrative)

Field	Content
Independent variable	Final pNPP concentration, 6 gradients: 0.05 / 0.10 / 0.20 / 0.40 / 0.80 / 1.60 mM (spanning the expected Km ≈ 0.25 mM)
Dependent variable	Initial rate v = ΔA405/min, taken as the slope of the 0–90 s linear segment; n = 3
Controlled variable	25 °C ± 0.5; pH 9.8 DEA buffer (prepared on the day); final enzyme concentration 0.1 U/mL; the same UV-2600 instrument
Control group	(1) blank (no enzyme); (2) heat-denatured enzyme (65 °C, 10 min); (3) previous batch standard ALP-2604 run alongside in the same batch
Replication and randomisation	Three replications per point, solutions prepared independently; instrument run order randomised; a single operator (Engineer Wu)
Three-level remove	Enzyme / substrate = essential part; DEA buffer / stop solution = function-degraded part; marker dye = non-essential part (not added)

Table 5-4 S3 output · SOP-ALP-042 v2.1 (illustrative abridged version)

Step	Action (parameter · tolerance)	Move	Key control points
1	Prepare DEA buffer at pH 9.8 (± 0.05), prepared on the day	Function generation	If pH is out of tolerance, remake the whole batch
2	Dilute the pNPP stock into six gradients (volume ± 1%)	Logic generation	Two-person check of the gradient labels
3	Pre-heat for 5 min at 25 °C, changed from 37 °C (± 0.5 °C)	Function generation	Do not start the run until the temperature is reached
4	Add enzyme to start the reaction; read A405 every 15 s for 6 points	Analysis	Start timing ± 2 s
5	Take the slope of the 0–90 s linear segment to obtain v	Analysis	Accept only if $R^2 \geq 0.99$
6	Run the blank, the heat-denatured control and the previous batch standard in the same way	Reasoning	Controls and samples on the same day and the same instrument
7	Enter raw values into the Data Type table; complete the four-gate columns on the same day	Continuous learning	Raw values read-only

Example | Table 5-5 S4 output · Data Type DS-2610-07 (illustrative data)

Measurement matrix (initial rate v, ΔA405/min, n = 3, mean ± SD):

[S] = 0.05 mM: 0.083 ± 0.004 | 0.10 mM: 0.143 ± 0.005 | 0.20 mM: 0.222 ± 0.007

[S] = 0.40 mM: 0.308 ± 0.008 | 0.80 mM: 0.381 ± 0.009 | 1.60 mM: 0.432 ± 0.010

∩(T) four gates: linearity R^2 = 0.997; within-point CV = 1.8%; blank drift = 0.004 ΔA/min; dilution back-calculation R^2 = 0.999 → all passed.

Traceability: UV-2600#0412, CAL-2603, operator Engineer Wu, 2026-10-14 09:20–10:05.

Figure 5-3 (illustrative data) Figure evidence of the Verdict Type: 18 initial rates (n = 3) on the M-M fitted curve

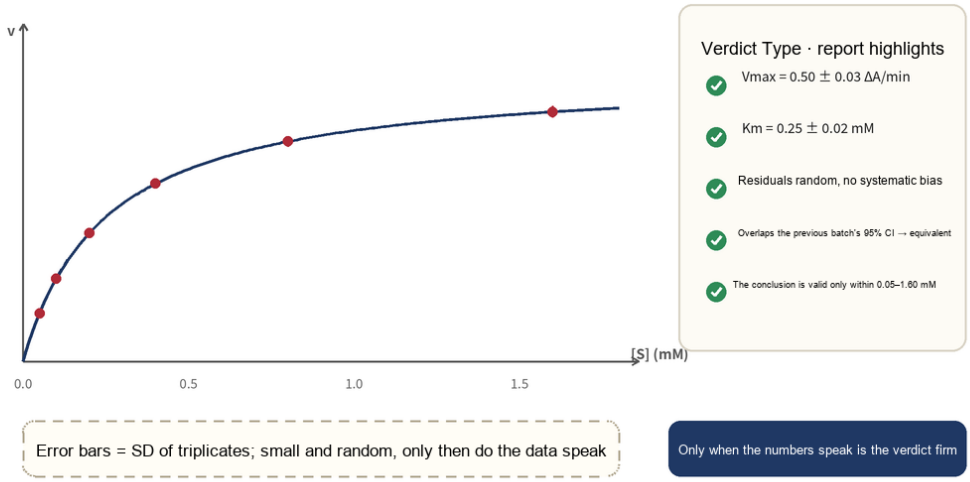


Figure 5-3 Graphical evidence for the Verdict Type: 18 initial-rate values pinned onto the fitted curve (illustrative data)

5.4 The Verdict Report and the Evolution Log

The S5 verdict pins the data onto the skeleton: Michaelis–Menten non-linear regression as the primary method, the double-reciprocal plot as a supporting check, yielding point estimates and 95% CIs for Km and Vmax, which are then compared with the previous batch specification. Once the comparison passes, S7 writes the conclusions and lessons of the whole round into the evolution log — and that is the starting point of the next round.

Table 5-6 S5 output · Verdict Type report VR-2610-07 (illustrative)

Field	Content
Skeleton and method	v = Vmax[S]/(Km+[S]), settled by non-linear regression; Lineweaver–Burk used only for inspection (six points collinear, R^2 = 0.998)
Parameters and uncertainty	Vmax = 0.50 ± 0.03 ΔA405/min; Km = 0.25 ± 0.02 mM (95% CI, illustrative)

Field	Content
Residual check	Residuals scattered randomly with no systematic shift; no outliers requiring handling
Scope	Valid at [S] 0.05–1.60 mM, 25 °C, pH 9.8; not extrapolated beyond this range
Comparison verdict	Previous batch specification Km = 0.25 ± 0.02 mM, Vmax = 0.50 ± 0.03 → CIs overlap, equivalence holds → hypothesis passes

Table 5-7 S7 output · Evolution Type log EV-ALP-003 (illustrative)

Field	Content
Version narrative	SOP-ALP-042 v2.1 → v2.2 proposal: this round passed all four gates and equivalence holds; v2.2 will incorporate a temperature-gradient stage
Lesson log	(1) blank drift once exceeded the limit → buffer now prepared on the day; (2) operator timing spread of ± 4 s → dual-person timing plus a spoken start cue
Next-round hypothesis	Temperature dependence of Km at three temperatures, 25 / 30 / 37 °C (Arrhenius inspection), together with determination of the inhibitor Ki
Trigger criterion	A new round is triggered when the parameters of two consecutive batches fall outside the previous batch's CI, when a new instrument is introduced, or when the SOP has gone six months without review

Fig. 5-4 Evolution Type · batch-to-batch evolution: every round the SOP grows a little

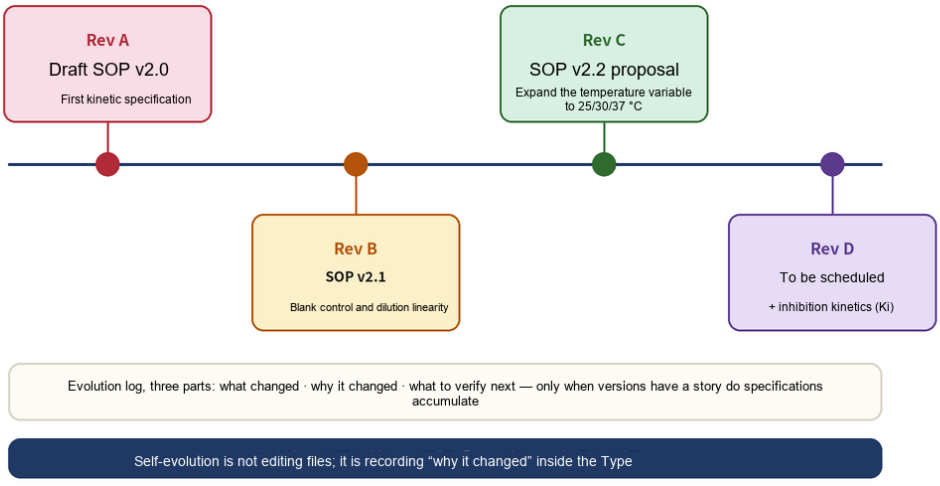


Figure 5-4 Evolution Type · inter-batch evolution log: every round of experiments ages the SOP by a year

5.5 Report Verification and Rules of Use

Before the S6 integrated report is handed over, three verification passes are made: numerical consistency (abstract = body = data table), evidence-chain completeness (station-by-station cross-check of the six stations), and completeness of declarations (the three items: illustrative data, range of applicability, disclaimer). Sign only once verification

has passed. In daily use, keeping the six rules beside the prompts is the safest arrangement.

Table 5-8 Rules for using the prompts (six rules)

#	Rule	Why
1	Open one stage at a time; do not open the next stage before the previous one is accepted	The seven stages are a relay, not a buffet; skipping a stage means skipping a gate
2	Every stage carries its criteria (∩(T) written into the closing)	An output without a referee turns acceptance into an argument
3	Every stage leaves feedback (that stage's three self-check questions)	Self-checking is the cheapest quality insurance
4	Example data are marked 'illustrative'	Illustrative passed off as real means trust bankruptcy
5	Moves must be named (e.g. Search + Reasoning)	Only when you know which move you used can you switch to the next one
6	Revisions must be logged (version number + date)	An evolution log without versions has no story

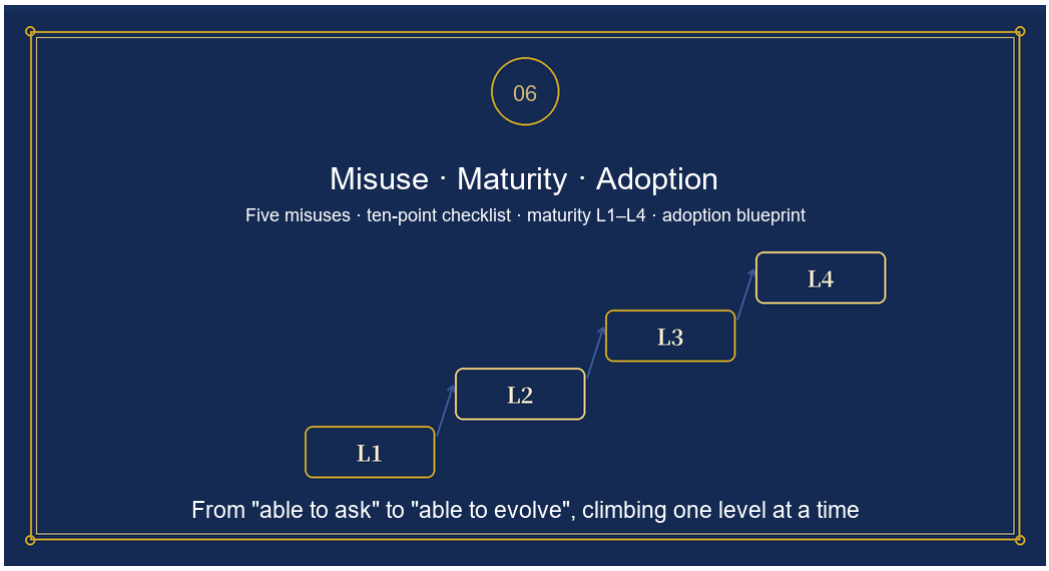
Chapter highlights

The full prompt text runs to seven stages in one consistent format: task, input, output, criteria, three self-check questions — copy it directly and use it once modified.

Engineer Wu's case rehearses all six Types end to end: hypothesis card → experiment design document → SOP → data table → verdict report → evolution log.

Every number is illustrative; what is genuinely valuable is the path of 'set the criteria first, then collect the data'.

6 Misuse, Maturity, Adoption



The final chapter is about lasting stability: it first lays out the five most common misuses, then gives you a ten-question checklist as a pre-flight check, and finally uses maturity levels L1–L4 and an adoption blueprint to settle the method into the laboratory's daily routine.

6.1 The Five Misuses

The misuses share three features: skipping the criteria, omitting the control, and inflating the conclusion. None of the three makes the data look better; they only make the conclusion more dangerous. On the left is the scene of the stumble, on the right the correct posture.

Figure 6-1 Five misuses: the left column is the pitfall, the right the correct practice

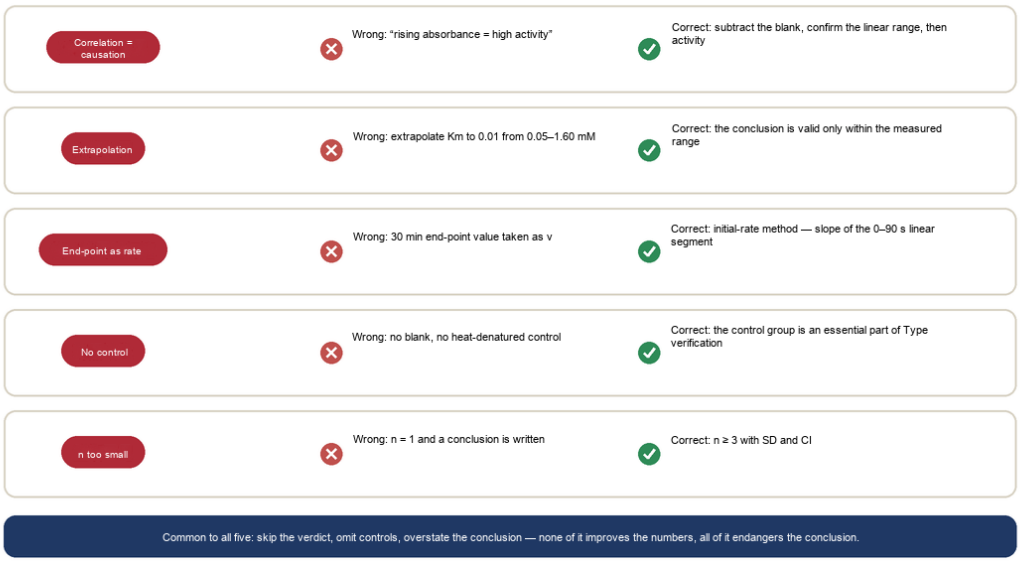


Figure 6-1 The five misuses: the stumble on the left, the correct posture on the right

Table 6-1 The five misuses compared

Misuse	Symptom	Correct response
Treating correlation as causation	'Absorbance rises = activity is high', concluded directly	Subtract the blank first, confirm the linear range; causation must be supported by controls and mechanism
Extrapolation beyond the range	Extrapolating from 0.05–1.60 mM data down to 0.01 mM	Conclusions hold only within the measured range; the applicable range must be stated
Endpoint method masquerading as initial rate	Reading the 30 min endpoint as the reaction rate	Initial-rate method: only the slope of the 0–90 s linear segment is v
Missing control	No blank, no heat-denatured control	The control is a necessary component of the Experiment Type; remove it and identity is lost
Insufficient replication	Drawing conclusions from n = 1, reporting only the mean and no SD	n ≥ 3, report SD and CI; investigate outliers in the experiment first

6.2 The Ten-Question Checklist

Before opening a case and before issuing a report, run through these ten questions. If three or more cannot be answered, hold off — this is not delay, it is time saved on rework.

Table 6-2 The ten-question checklist (pre-flight check)

#	Question	Maps to
1	Is the hypothesis falsifiable? What data would overturn it?	Hypothesis Type

#	Question	Maps to
2	Are the independent, dependent and controlled variables clearly separated?	Experiment Type
3	Are the controls complete? Blank / positive / reference material?	Experiment Type
4	Are the number of replications and the randomisation planned?	Experiment Type
5	Is the linear range for the initial rate confirmed? (Does R^2 clear the threshold?)	∩(T) four gates
6	Have all four ∩(T) gates passed? Is there a sign-off?	∩(T) four gates
7	Which equation is the reasoning skeleton? Has it been committed to first?	Reasoning skeleton
8	Does the conclusion trace back to the hypothesis? Are all six stations cleared?	Type evidence chain
9	Does the conclusion go beyond the measured range?	Verdict Type
10	Are the next round's hypotheses and lessons written into the evolution log?	Evolution Type

6.3 Maturity L1–L4

Maturity is not a job title; it is the degree to which outputs can be acceptance-checked. A four-rung ladder: L1 Improvisation (ask whatever comes to mind), L2 Template (six-Type cards and S1–S7 become routine), L3 Criterion (∩(T) acceptance is automatic, the evidence chain takes shape), L4 Evolution (the evolution log closes the loop, the SOP revises itself). Each level rests on the one below it; skipping a level yields only the illusion of looking accurate.

Figure 6-2 Maturity L1–L4: the four-rung ladder from "able to ask" to "able to evolve"

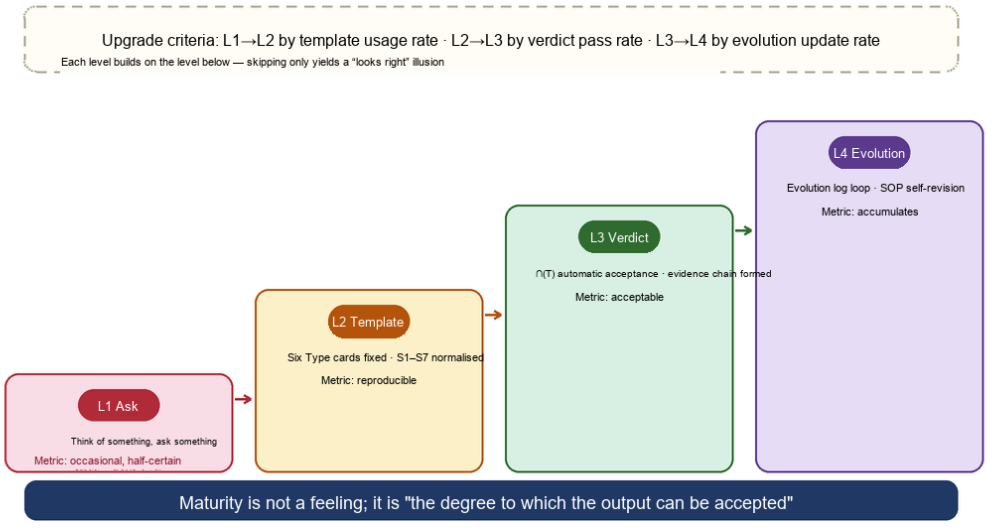


Figure 6-2 Maturity L1–L4: the four-rung ladder from asking well to evolving

Table 6-3 Maturity L1–L4 compared

Level	Characteristics	Upgrade criterion	Common sticking point
L1 Improvisation	Prompts follow whatever comes to mind	Template usage > 60%	Outputs not reproducible, reporting conventions inconsistent
L2 Template	Six-Type cards and S1–S7 become routine	Criterion pass rate can be measured	Templates exist, but criteria still rely on feel
L3 Criterion	∩(T) acceptance is automatic, the evidence chain takes shape	The evolution log is updated every month	Acceptance is strict, but nothing grows
L4 Evolution	The evolution log closes the loop, the SOP revises itself	(sustain continuously)	Complacency, review stops

6.4 Adoption Blueprint

Adoption does not need a big bang. Three months, three milestones: in month one, pilot a single familiar protocol (a routinely run quantitative enzyme assay is the recommended choice); in month two, roll the six-Type cards and criteria out to the whole group; in month three, bring ∩(T) acceptance and the evolution log into routine review. Pilots succeed most often when the experiment is routinely run, quantifiable and backed by data from earlier batches.

Table 6-4 Three-month adoption blueprint

Milestone	Action	Output	Acceptance
Month 1 · Pilot	Take one familiar protocol through all seven stages	One complete set of six-Type documents	All four gates passed + report signed
Month 2 · Roll-out	Turn the six-Type cards and criteria into templates; train the whole group	In-group template library	Template usage > 60%
Month 3 · Routine	Bring ∩(T) acceptance into the regular meeting; open the evolution log	Acceptance checklist + evolution log	At least one evolution record per month

Chapter highlights

The common antidote to all five misuses: skip no criterion, omit no control, inflate no conclusion — do none of these three things.

Use the ten-question checklist as a pre-flight check; if three questions cannot be answered, hold off.

Climb maturity L1–L4 one rung at a time; adopt through a three-month pilot roll-out, not a big bang.

Review: Three Takeaways

From start to finish this book does one thing: it turns "accurate" from an adjective into a method. Before you leave, take three things with you.

Figure R-1 Three takeaways: the Type + Number replica, the key to the six Types, the four-layer gate

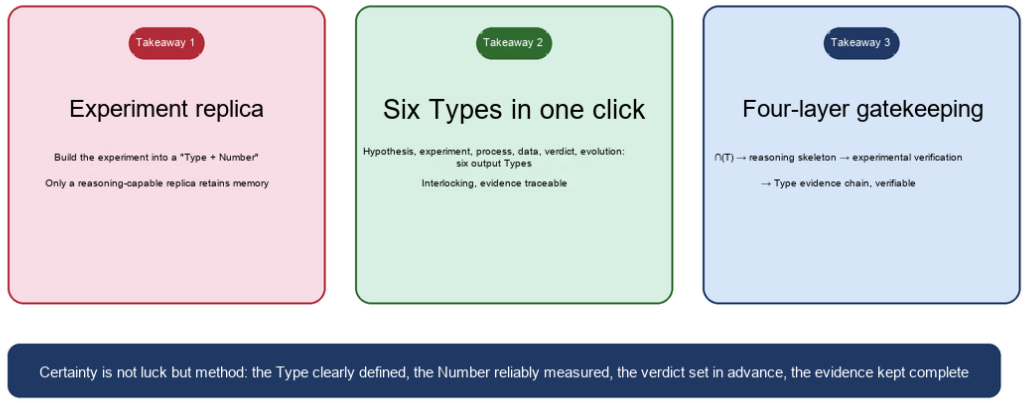


Figure R-1 Three takeaways: the Type + number twin, the chain of six Types, the four-layer gate

Takeaway 1 · The experimental twin: Type + number

Build the experiment into a reasoning-capable twin of "Type + number": record structure as Type (Container, Composition, Abstract) and measurement as number (with the full three-part traceability set).

While the twin exists, the memory exists — a different person, a different batch, half a year later, the reasoning path can still be traced.

Takeaway 2 · One chain of six Types: hypothesis, experiment, protocol, data, verdict, evolution

The hypothesis is falsifiable, the experiment is discriminable, the protocol is reproducible, the data is recomputable, the verdict is acceptance-checked, the evolution is continuable — the six Types interlock and the evidence traces back.

The full S1–S7 prompt texts are in Chapter 5; copy them straight, adapt them and use them.

Takeaway 3 · Four layers of gatekeeping: accuracy that can be checked

∩(T) criterion → reasoning skeleton → experimental validation → Type evidence chain: none of the four gates may be skipped; space × time cross-acceptance gates on both axes.

Accuracy is not luck, it is method: agree the criteria first, and only then dare to sign the conclusion.

Table R-1 Master list of tools in the book (exit checklist)

Tool	Chapter	Use in one line
The three brackets ( ) { } [ ]	Chapter 2 (Output Skeleton)	Container identity, formulation composition, category criteria — build these first when opening a case
The three levels of the remove criterion	Chapter 2 (Output Skeleton)	Remove one part to see the whole: necessary component, function-degraded component, non-essential component
∩(T) four gates	Chapter 3 (Accuracy Gatekeeping)	Linearity, replication, blank, dilution — only when all pass may you fit
Reasoning skeleton	Chapter 3 (Accuracy Gatekeeping)	Commit to the equation first, then let the data speak
Four-step framework + six steps	Chapter 4 (Operating Procedure)	Clarify the requirement first; walk the six-step loop
Three-layer judgement	Chapter 4 (Operating Procedure)	A three-question scan: Is it? Does it have? Is it right?
S1–S7 prompts	Chapter 5 (Prompt Examples)	A seven-stage relay, one stage at a time
Ten-question checklist	Chapter 6 (Governance and Adoption)	Pre-flight check; if three questions cannot be answered, hold off

Appendix A The Toolbox

A.1 Terminology Quick Reference

Table A-1 Terminology quick reference (thirteen rows)

Term	Definition (used only, not taught)	Application point in this book
Type	The smallest identifiable unit of structure	Every output in the book is named with Type, never Form
Space Type	The Type that describes "what it is": Container, Composition, Abstract	Sample, formulation, experiment category
Time Type	The Type that describes "how it is done": verb = function = move	SOP actions, analytical methods, evolution
Container ( )	C = (id(C), S(C)); identity does not change with the contents	Well, tube, petri dish
Composition { }	BOM and the three-level remove criterion	Reaction formulation, culture medium
Abstract [ ]	T = (F(T), ∩(T)); open rules define the category	Experiment category, specification
∩(T) intersection criterion	Membership only when every feature passes	The four data-quality gates
remove criterion	Take it off: does it still recognise itself?	Necessary component / function-degraded component / non-essential component
id(C) identity	A marker that does not change with the internal space	Well number, sample barcode
Four-step framing	Distinguish similarity and difference → trigger abstraction → name the dimensions → assign membership	Requirement clarification
Three-layer judgement	The three questions on Container, BOM and Abstract (Is it? Does it have? Is it right?)	Experiment eligibility review
Reasoning skeleton	Commit to the equation first, then let the data speak	Michaelis–Menten
Type evidence chain	A six-station chain tracing the conclusion back to the hypothesis	Experimental records and sign-off

A.2 The Fourteen Moves × Their Application in Biological Experiments

Table A-2 Complete table: the fourteen moves on the move board × their application in biological

experiments

Group	Move	Application in biological experiments
Search	Type–time function search	Look up literature methods, standard methods (e.g. pharmacopoeia), previous-batch SOPs
Search	Design search and extraction	Extract the experimental design from the literature: gradient, controls, number of replications
Generation	Design patterns	Apply patterns such as control design, factorial design, blocked design
Generation	Logic generation and application	Derive variable logic, criterion logic, conditional branches
Generation	Function generation and application	Generate SOP actions, data tables, analysis-script specifications
Thinking	Problem-solving	Diagnosing experimental failure: troubleshooting failed gates and abnormal controls
Thinking	Analysis	Data analysis: fitting, residuals, outlier determination
Thinking	Design	Experimental design: gradient layout, sample-size estimation
Thinking	Reasoning (cause→effect / effect→cause / simulation / statistical / reverse…)	Inferring mechanism, separating correlation from causation, verifying conclusions in reverse
Thinking	Thinking methods (forward / reverse / lateral / vertical / first principles…)	Rethink the hypothesis from another angle, take anomalies apart from first principles
Thinking	R&D and exploration	Exploring new indicators, new methods, new inhibitor conditions
Evolution	Continuous learning	Batch-to-batch learning: fold this round's data into the specification baseline
Evolution	Reconstruction and recombination	SOP reconstruction: merge steps, tighten tolerances
Evolution	Self-evolution	Self-revising specifications, next-round case opening driven by the evolution log

A.3 Notation Quick Reference

Table A-3 Notation quick reference

Notation	Read as	Used for
T = (F(T), ∩(T))	Category = feature set + intersection criterion	Defining experiment categories, the four-gate criterion
C = (id(C), S(C))	Container = identity + internal space	Wells, sample containers
remove(p,M)	Remove part p from the whole M	Three-level classification of a formulation BOM
remove(p,M) ≠ M	Identity is lost after removal (necessary component)	Enzyme, substrate

Notation	Read as	Used for
remove ≡ M' ⊂ M	Function is degraded after removal (degraded component)	Buffer, stop solution
remove(p,M) ≡ M	Unaffected after removal (non-essential component)	Labelling dye
v = Vmax[S]/(Km+[S])	Michaelis–Menten skeleton	Reasoning skeleton, non-linear regression
1/v = (Km/Vmax)(1/[S]) + 1/Vmax	Double-reciprocal inspection form	Lineweaver–Burk plot
S(t+1) = g(S(t), move, event)	The state evolves with the moves	Evolution log, SOP version drift

> **Final reminder**
>
> All data and document numbers in this book are illustrative; in real use, follow each laboratory's own regulations, safety rules and quality-control system.
>
> The English term is Type, not Form — take that convention with you as well.

So that every experiment,

has a reasoning-capable Type.

From hypothesis to evolution,

accurately verifiable.

MIND AI × ONTYPOLOGY

AGI · The Eternal Way | Unified knowledge · unified thinking · unified wisdom